Engineered Intelligence: Tracing Computational Design from Search to Evidence

A Retrospective Study of Representation, Search, and Assessment

Research draft prepared for Jepson Taylor | Human author approval pending

Abstract—Automated design searches can retain extensive candidate records while leaving the evidence behind a selected result difficult to interpret. We present Jarambit as a retrospective study linking an application-specific geometric representation, recorded search observations, and candidate-specific numerical evidence. The largest archive contains 1.95 million search-evaluation records; historical genealogy and intermediate screening are analyzed separately because their scoring and retention regimes differ. Three findings delimit what these records establish. Evidence coverage is uneven: extensive genome and score tables do not imply retained parentage or downstream assessments. Exact content joins recover proposal occurrences without recovering parents. In the historical run, the highest stored search score fails an earlier Boolean predicate that a lower-ranked candidate satisfies; this is distinct from later qualification. For a separately assessed export, selected solid-model observables agree closely between implementations on a shared mesh, while separate plate-refinement sequences do not admit the chosen extrapolated-error estimates. The contribution is an inspectable connection between representation, candidate identity, assessment stage, and source artifacts, with incompatible scores and missing observations kept explicit. These distinctions make the reported evidence easier to question and reconstruct. An unsupported historical buckling inference is withdrawn while its original record is retained. The findings remain computational and retrospective; physical performance and general search advantage are unestablished.

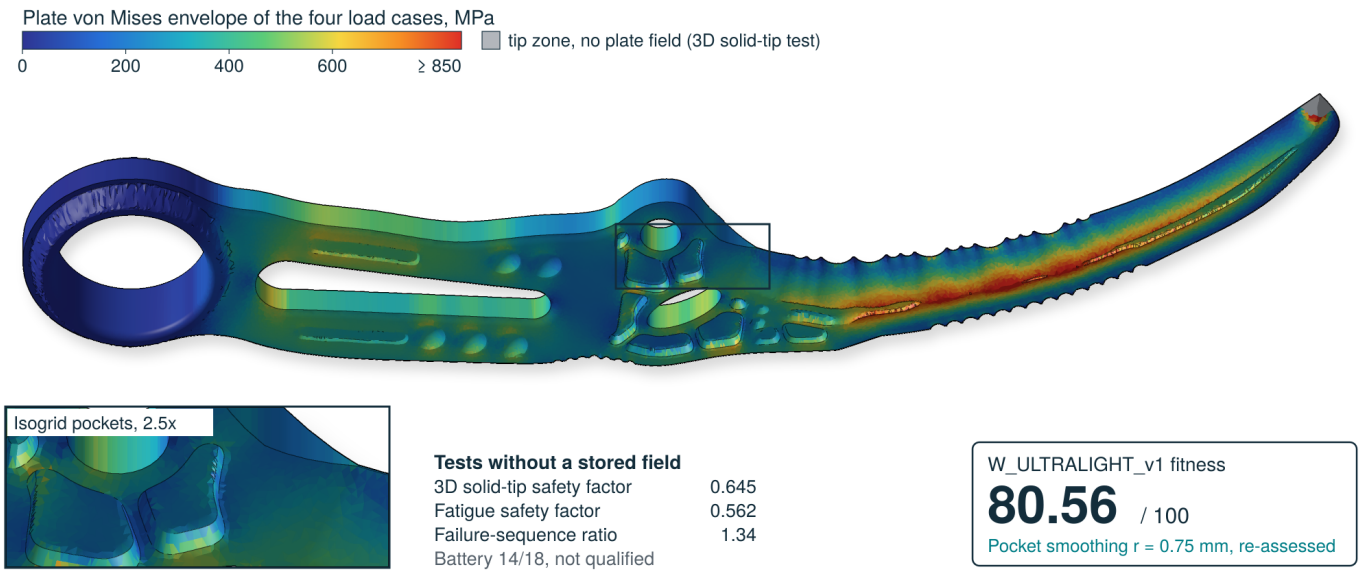


Fig. 1. Best evolved design by W_ULTRALIGHT_v1, ec3i042_r01_p10_c56, after surface-tension pocket smoothing ($r = 0.75$ mm; Section VI): re-assessed fitness $F = 80.561227$ of 100, against $F = 80.623796$ as exported (variant A). The smoother runs inside the CAD export of the same genome, so the smoothed design is a new STEP solid and every objective is measured again; it keeps the 5-point CAD/B-rep objective. 3D mass 83.56 g, battery 14/18, not qualified. Same camera, lighting, 2.5× isogrid-pocket inset and MPa colour scale as variant A; colour is the plate-model von Mises envelope stored by this re-assessment, grey the 4 mm tip zone that only the 3D solid-tip test covers.

Index Terms—Engineered Intelligence, computational design, evolutionary computation, executable genomes, CAD, quality diversity, finite elements, evidence provenance, reproducible populations.

Research draft; author approval pending. Retained computational evidence, with physical observations not reported in this study. Source and supplement accessions are described in Section XII.

I. Introduction: Tracing Search to Evidence

JARAMBIT studies computational design for a titanium hooked utility knife with an integral ring. Its largest admitted cohort contains $1{,}954{,}560$ evaluated birth records across ten completed runs. Smaller historical and screening collections supply different observations, and separately assessed exports connect particular geometries to numerical records. A candidate's search context and its assessment history therefore need to be read together.

We use *Engineered Intelligence* for a process connecting a versioned representation and decoder, recorded variation and selection, declared objectives, and inspectable observations. Jarambit's documented schema has 128 numeric coordinates, including conditional and discretized parameters, plus a categorical material field. A genome, a contextual search score, and an assessed artifact each describe a different part of that process.

A. Research Question and Three Findings

The study asks: *what can the retained links establish about a computational design result?* Three findings organize the answer:

1) **Coverage depends on the observation stage.** Extensive genome and score tables coexist with selective parentage, geometry and later-assessment records. The historical genealogy, native populations and intermediate screens retain different kinds of evidence.
2) **Search rank and a recorded assessment predicate describe different outcomes.** In historical Run 1, the highest-scoring birth, Individual 1095, fails that run's recorded certification predicate, while lower-ranked Individual 1543 meets it. This categorical comparison is scoped to the earlier predicate; later all-requirement qualification is a separate assessment.
3) **Numerical agreement and error estimation answer different questions.** Selected solid-model observables agree closely between two implementations on a shared mesh. Separate plate-refinement sequences refuse the chosen extrapolated-error estimates. Each result carries its own inputs and inference scope.

B. Contribution and Reading Guide

The contribution is a source-linked case study joining representation semantics, candidate identity, score context and candidate-specific observations in one inspectable account. Recorded ancestry illustrates reuse within the fixed representation. Coverage views distinguish available observations from missing ones, and the numerical example separates shared-input consistency from mesh-study admission. These concrete links let a reader trace what a displayed result means without treating a collection of scores as a uniformly assessed population.

Section III defines the records and counting units; Sections IV–VII explain representation and inheritance. Section IX develops the findings, and Section XI interprets the numerical evidence. The analysis is computational and retrospective; Section XIV consolidates its scope and limits.

II. Background and Related Research

A. Computational Design and Representation

Computational design already includes mature approaches to changing structural form. Sigmund and Maute review density, level-set, evolutionary, and other topology-optimization approaches [13]. Jarambit's documented native schema instead varies an application-specific parameterization under a fixed decoder. Its coordinate count is not evidence of an unrestricted topology search or a newly discovered representation language.

DeepCAD learns CAD-operation sequences for autoencoding and generation [2]; Text2CAD generates parametric CAD sequences from text and reports visual, parametric, and geometric evaluations [3]. These representation choices differ from Jarambit's numeric schema. None supplies a common mechanical benchmark for the present study.

Evolutionary engineering also predates language-model assistance. The cited 2003 ST5 report describes evolved antenna representations, fabricated prototypes, and comparison with a hand-designed antenna; flight qualification was still pending in that report [4]. MAP-Elites retains representatives across user-defined behavioral dimensions [5]. Such archive selection differs from retaining all observed births, and archive occupancy alone does not measure population diversity. Jarambit's cohort-specific archive and score definitions therefore remain part of its reported context.

B. Research Provenance and Artifact Access

The closest context for this study's evidence links is research provenance. PROV-DM distinguishes entities, activities, derivations, responsible agents, and collections [14]. These concepts help explain why a candidate record, a search activity, an exported geometry, and a reporting derivative require distinct identities. Jarambit's retained links are compared on these dimensions; no PROV conformance or general interchange implementation is established here.

Implemented provenance systems provide a closer comparison than metadata principles alone. AiiDA stores calculations, workflows, inputs, and outputs in a traversable provenance graph and supports automated computational workflows [18]. Workflow Run RO-Crate packages workflow-execution provenance with associated inputs, outputs, and code across execution systems [19]. These systems already address provenance capture and exchange.

Jarambit contributes a retrospective case analysis at the boundary between records and published engineering claims: which candidate and observation stage a quantity describes, which denominator a count uses, and which evidence remains unavailable. The present study does not benchmark its packaging against AiiDA or RO-Crate, or establish superior capture, interoperability, or reader comprehension.

The FAIR principles address discovery, access, interoperability, and reuse of research objects [15]. Software citation principles additionally distinguish persistent identification, access, and specificity of the software version used [16]. A hash identifies bytes; meaningful reconstruction also requires retrievable artifacts and interpretation of their types, revisions, and relationships. This study's contribution is a concrete

retrospective account of those relationships and gaps, rather than a new general provenance model.

We cite the inspected Jarambit publication baseline as a source snapshot [1]. It is distinct from the cohort producers and from the final manuscript release accession (Section XII). A methods publication, a source snapshot, an executed binary, and a generated figure likewise denote different objects.

C. Numerical Credibility and Visual Interpretation

NASA's models-and-simulations standard supplies context for separating credibility questions; compliance is not claimed [7]. The scikit-fem and DOLFINx methods references identify finite-element software roles, while execution provenance must identify the versions and model inputs actually used [8], [9]. Selected observable agreement on a shared mesh and admissibility of a grid-refinement estimate answer different questions (Section XI).

Quantitative views also need explicit units, transformations, masks, and legends. Crameri *et al.* explain how uneven color gradients can introduce visual distinctions absent from the data [17]. A percentile summary therefore requires its own interpretation and cannot inherit the physical units of its component fields. The paper's spatial views are reporting devices, not a separately validated visualization method.

III. Study Design and Retained Evidence

This is a retrospective study of selected retained records, rather than a prospective comparison of optimizers. We ask what those records establish about representation, recorded search preference and later numerical assessment. The three cohorts have different scoring and retention policies; the separately assessed exports supply additional candidate-specific observations, rather than downstream measurements for every search birth. Figure 2 separates these retained evidence paths. Individual 2000 is a stable publication alias for a retained export; the integer does not state its rank, birth sequence or generation. Exact certificate and genome hashes link it to a retained proposal, while its per-child parent identifiers remain unrecorded.

A *birth* is a recorded candidate-creation event. A *membership* is an appearance in an island slot, including retention or migration of an existing individual. An *archive representative* is a selected record under its archive's objective and descriptor context. An *assessed export* has a specifically bound artifact/assessment bundle. These are different counting units: one genome can occur in several births or memberships, and a certificate identity alone does not establish that its CAD or assessment artifact is retained.

Sources H and P are the historical reconciliation and proposal/screen join accessions; N is the completed generation-800 population accession. E is the automatic manifest plus original field/assessment/verification contracts; C is the controlled-export scorecard accession. Availability is local to these accessions. The native population tables retain all 128 numeric coordinates and material, but their size does not establish complete parentage, per-birth exported CAD, later solid assessment or independent physical replication. Their accession records an untracked producer state rather than a clean executed-source binding; five of the ten completed-run manifests have no process-configuration hash. The Supplement distinguishes that accession statement from the separately retained baseline and local-patch hashes.

A. Run Identities and Capture Overlap

The native inventory contains 18 current campaign directories: ten completed generation-800 runs and eight others. An earlier staged snapshot has eleven summary files, of which four are earlier captures of runs later included among the ten completed runs (public Runs 6–9). These are overlapping views of original run identities, not 21 independent runs. Nine aborted crescent attempts are a separate accession; seven retain 216 summary rows in total and two have no retained CSV rows. Summary rows describe generation/island aggregates, not new births. Capture/receipt times are kept separate from unverified historical process start/stop times.

B. Selection and Interpretation

The selection registry designates Individual 2000 as the illustrated assessed export because its retained bundle connects export geometry, plate fields, mesh-study observations, solid comparison and scorecard. Its recorded human editorial-authority category does not independently identify a named chooser. This illustration role confers neither championship nor qualification. Table VII separates it from source-bound stored score leaders and other assessed records. The historical leader illustrates a later-assessment disagreement; neither example is a random sample from the large native archive. This reporting reconstruction does not invent scores for missing observations or turn them into a measured qualification fraction. Historical weighted-score defaults are preserved and disclosed separately in Section VIII.

IV. Representation and Candidate Identity

A. A Gene Is a Semantic Contract

A gene requires an explicit coordinate index, physical or discrete type, unit, domain, coupling law, and decoder revision. A serialized floating-point token alone cannot distinguish a millimeter displacement from an angle, a feature-count selector, or an architecture switch. Those semantics determine what a stored genome means when it is decoded.

The documented native schema stores 128 numeric coordinates and one categorical material field. Some numeric coordinates express continuous parameters; others are interpreted through thresholds or discretization. The nominal coordinate count is therefore neither the number of independently active parameters nor a measure of geometric freedom. Field roles are summarized below; exact domains and decoding rules require the producer revision attached to each cohort. Historical-producer details are retained in Supplement I, without attributing them to every executed workflow.

TABLE I
CONTEXT BY REPRESENTATION, SEARCH, EVIDENCE, AND VALIDATION SCOPE. ROWS DESCRIBE DISTINCT RESEARCH PURPOSES RATHER THAN A PERFORMANCE RANKING.

Context	Representation	Search or transformation	Evidence emphasized	Validation scope
Topology optimization	Varies by formulation	Numerical design updates	Structural objectives and constraints	Method- and model-specific
DeepCAD / Text2CAD	CAD-operation sequences	Learned generation	Reconstruction or generated CAD	Reported geometric/parametric evaluations
Provenance systems	Workflow objects and relations	AiiDA capture; RO-Crate packaging	Execution history, artifacts, and derivation	Provenance/exchange scope, not physical validation
This study	Fixed CAD schema and candidate identities	Retained evolutionary and assessment records	Coverage, ancestry, scores, and selected numerical observations	Retrospective computational evidence

TABLE II
COVERAGE OF THE ADMITTED STUDY RECORDS. A: AVAILABLE; P: PARTIAL/SELECTED; NR: NOT RETAINED IN THE NAMED ACCESSION; U: COVERAGE NOT ESTABLISHED. PHYSICAL OBSERVATIONS ARE NOT REPORTED IN THIS STUDY (NPUB); THIS DOES NOT INFER THAT NO EXPERIMENT EVER OCCURRED. EACH ROW'S SOURCE KEY APPLIES TO ALL ITS CELLS; THE SUPPLEMENT SUPPLIES EXACT PATHS, HASHES AND QUALIFICATIONS.

Records (source)	**Denominator**	**Genes/ material**	**Parentage**	**Exported geometry**	**Spatial fields**	**Later assessment**	**Physical**
Historical genealogy (H)	1984 births; 2176 memberships	A	A within run	NR in cited accession	NR in cited accession	P: selected strict records	NPub
Proposal/screen campaign (P)	3649 proposals; 2259 screens	A in proposals	NR per-child IDs	P: selected exports	P	P: selected summaries/bundles	NPub
Ten completed native runs (N)	1,954,560 births	A	U beyond admitted tables	P: separate STL accession	NR per birth	NR per birth	NPub
Automatic assessed export (E)	Individual 2000; one assessed	A	NR per-child parents	A	A plate; NR full solid	A saved bundle	NPub
Controlled exports (C)	Individuals 2001/2007, separately scoped	A	U birth link	A	P	A saved scorecards	NPub

TABLE III
FUNCTIONAL GROUPS IN THE DOCUMENTED NUMERIC SCHEMA.

Group	Role and stored type	Conditional interpretation
Silhouette	Numeric normal offsets around a reference outline	Some stations can be pinned by the producer's bounds.
Planform and features	Dimensions, angles, ratios, and feature selectors	A stored numeric feature setting need not activate a feature.
Internal machining	Pocket, window, and perforation parameters	Feature counts and zone switches use discrete decoding rules.
Continuous-rail layer	Numeric architecture switch, dimensions, and thickness knots	The switch selects the thickness architecture; only the selected knot count contributes.

B. Coupling and Conditional Activity

Clamping, thresholding, and coupled bounds distinguish stored coordinates from the geometry they produce. In the bound historical producer, the continuous-rail switch is thresholded, the active knot count is floored and bounded, and enabling the layer disables several legacy features. When it is disabled, knot coordinates are set to stock thickness during clamping. The decoder and its configuration thus belong to the interpretation of a record, rather than being incidental implementation details. A numeric change can be inactive, discretized away, or constrained by another coordinate; changed-coordinate counts alone do not establish geometric change.

C. Birth, Genome, Membership, and Export

A birth identifies a recorded generation event; a genome serialization identifies stored coordinate and material values; a membership identifies a population-slot occurrence; and an assessed export identifies its own geometry and assessment bundle. These objects can share an individual identifier without sharing a denominator. The historical cohort, for example, retains repeated genome serializations and repeated appearances of earlier births. A display mesh, a plate mesh, and a solid submodel likewise represent distinct discretizations, each requiring its own source binding (Section XII).

The inheritance ledger counts changed numeric coordinates by exact numeric inequality on parsed CSV values. Its raw-coordinate L_2 norms mix units and discrete semantics. They are audit summaries of stored coordinates, not physical distances between shapes; the detailed edge ledger is retained in Supplement I.

V. OBSERVED MORPHOLOGY AND FEATURE PROVENANCE

The best evolved design by W_ULTRALIGHT_v1, ec3i042_r01_p10_c56 ($F = 80.623796$ as exported, 80.561227 after pocket smoothing, Figure 1), makes the representation tangible: its retained Stage-2 bundle, re-assessed with the current evaluator, connects a geometric object with its genome, exported STEP and STL, scorecard and solid-tip observations, and Figure 3

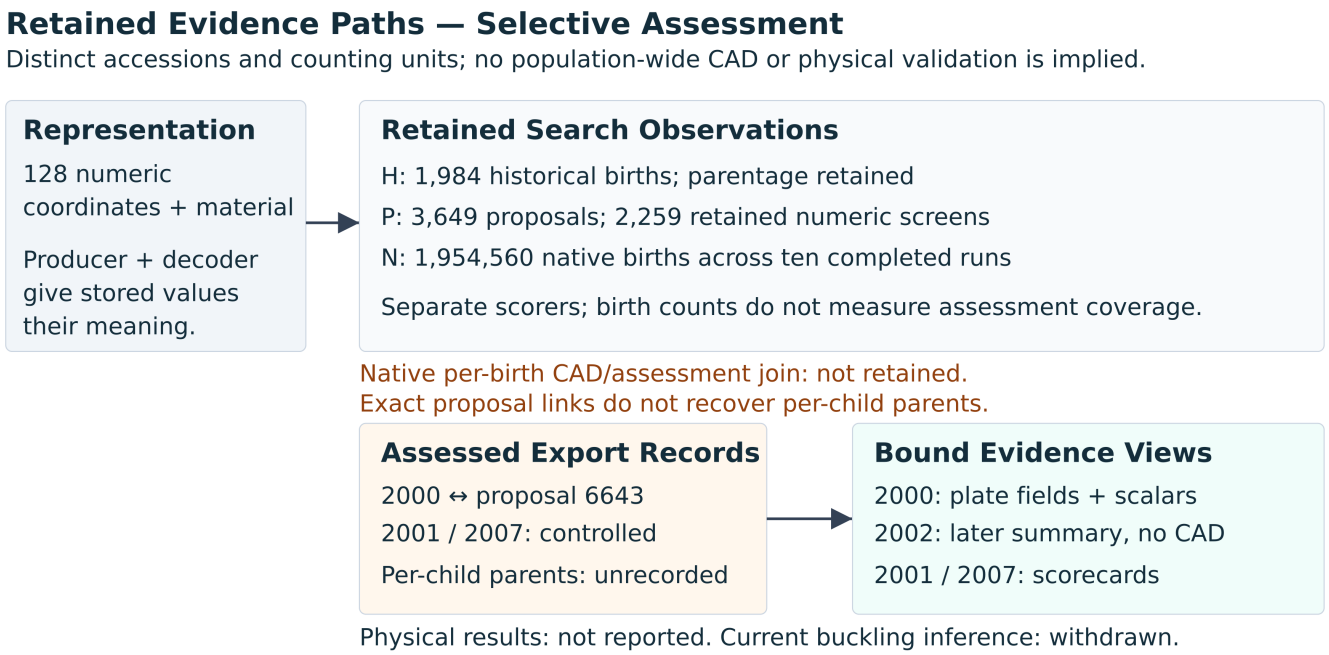


Fig. 2. Distinct retained evidence paths. Search observations and selected assessed exports have separate denominators and source bindings; a complete per-birth native CAD/assessment join is not retained. An arrow connects an assessed export only to its own retained evidence. Physical observations are not reported in this study.

shows that object in relief. The earlier illustrated example, Individual 2000, still supplies the retained plate response fields analysed in Sections VIII–XI. Three questions organize the interpretation: which feature families the representation makes available, which features an identified artifact actually contains, and what evidence establishes their consequences.

A. An Authored Feature Vocabulary

The documented representation provides silhouette and opening controls alongside longitudinal channels, face relief, patterned pockets, perforations and ring grooves. The producer's genome and phenotype sources define these families explicitly, including a blade fuller, triangular isogrid pockets, lattice features and ligament flutes; the reference source separately defines a ring-face groove[1]. These are authored parts of the vocabulary available to the decoder, rather than feature families whose invention is demonstrated by the illustrated export.

This vocabulary gives ordinary names to otherwise abstract coordinates. A fuller names a channel family; a pocket names a recessed region; a perforation names a through-opening. Material remaining between openings can be described as a bridge without assigning it a particular structural cross-section. Conditional switches, coupled bounds and construction rules determine how stored settings become geometry (Section IV). The available vocabulary and an individual feature's realization are therefore distinct observations.

B. Occurrence Requires an Identified Geometry

An occurrence claim belongs to a particular export and decoder context. The appropriate evidence is the retained geometry with its candidate identity and source binding. A nonzero gene setting alone does not establish the realized shape. Likewise, the boundaries in a response-colored image identify changes in a displayed field; they are not measurements of a geometric cross-section.

Neutral geometry and response views should consequently be read together, with a consistent orientation and an explicit account of the pictured region. Additional local anatomy callouts require a bound geometry asset and crop transformation. Until those bindings are supplied, descriptive names remain pointers for inspection rather than authenticated measurements of the highlighted regions. A content match can identify a recorded occurrence; a nearby genome does not establish a parent or the history of an individual feature.

C. Utility Is a Separate Engineering Claim

The saved assessments concern the illustrated candidate under their stated models and loads. They do not isolate the contribution of one channel, pocket or opening. This section therefore describes the representation and its inspectable artifacts without assigning feature-specific mass savings, efficiency, optimality or manufacturing qualification. Its value is explanatory: readers can connect the numeric representation to a concrete object while keeping feature availability, observed occurrence and demonstrated utility separate.

VI. Surface-Tension Pocket Smoothing

Why the walls were chipped. The exported solid is a height field over a fixed planform: faces $z = \pm t(x, y)/2$, with side walls on the outline and on every through-hole. Every weight-saving feature (isogrid pockets, lattice dimples, fullers, flutes, relief) is a removal depth $D = t_{\text{base}} - t$. Unsmoothed, that

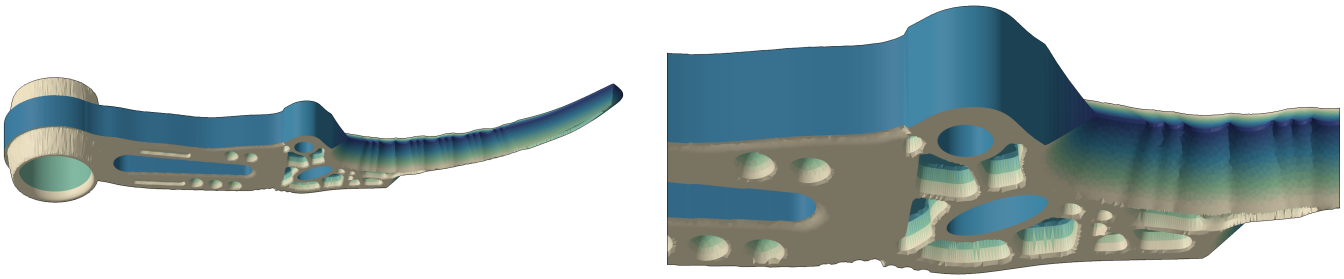


Fig. 3. The best evolved design, ec3i042_r01_p10_c56, after surface-tension pocket smoothing at $r = 0.75\,\mathrm{mm}$ (Section VI; the as-exported relief is level L0 of the smoothing ladder), in relief: whole design (left) and a close-up of the guard and pivot zone (right) from one oblique camera under a low raking light, with the thickness direction exaggerated $2.5\times$ and faces coloured by height above the mid-plane (dark blue at the mid-plane to cream at the $2.62\,\mathrm{mm}$ stock half-thickness; the finger ring saturates). Visible are the finger ring, the handle slot, two-sided isogrid pockets around the pivot and guard (full-thickness ribs between floors $1.35\,\mathrm{mm}$ deep on each face, an I-beam-like section), a pierced-fuller window, rows of relief pockets along the handle, the ground blade bevels and the edge serrations. Its ancestor w406t3_r01_p10_c16 (80.299390) has the same pocket layers; 19 of 129 genes differ, mostly slightly. Same renderer and settings as level L0 of the smoothing ladder (docs/paper/smoothing-ladder.pdf).

depth reaches the CAD through the $0.25\,\mathrm{mm}$ thickness raster by nearest-cell lookup, so most pocket walls are vertical steps along a staircase. A constrained Delaunay triangulation (CDT) of about $0.7\,\mathrm{mm}$ triangles cuts across those steps, so the walls render as chipped (Figure 4b, left). Mesh smoothing (docs/paper/smoothing-ladder.pdf) cannot remove this: it pins every vertex on an edge sharper than $30°$, which is where the chips are.

Method. We smooth the removal depth itself, before meshing, as a liquid would. Each pocket is a basin: a connected component of $D > 0.02\,\mathrm{mm}$ (a watershed), merged across gaps narrower than $r/2$. Inside its basin the depth evolves for time $T = r^2/2$ by level-set curvature motion with a per-basin pressure,

$$D_t = |\nabla D|\,\nabla\cdot\frac{\nabla D}{|\nabla D|} + \lambda_b\,|\nabla D|, \tag{1}$$

where λ_b keeps the removed volume of basin b constant (the volume of oil in the pit). Every depth contour then shortens like a two-dimensional droplet of fixed area under line tension: chips and spikes, which have the highest curvature, go first; straight rib edges stay straight and corners round to a radius of about r; and walls keep their steepness, so ribs keep their width. No basin may end deeper than its own deepest original point, so the volume constraint widens a thin groove instead of drilling it toward the minimum web. The planform, through-holes, grind faces, bore collar, raised features, brand pad and a $1\,\mathrm{mm}$ rim of unpocketed material along the outline and holes never move.

The flow runs on a $0.125\,\mathrm{mm}$ grid (2–3 s per design); the exporter reads $t = t_{\mathrm{base}} - D$ by bilinear interpolation and adds CDT vertices in three rows along every wall (depth contours at 8, 50 and 92 % of the local wall height) and in pairs straddling the grind boundary, so triangle edges follow each wall instead of cutting across it. Equation (1) is the $a = 0$ case; $a = 1$, a liquid-film flow, is compared below. The switch is JARAMBIT_POCKET_SMOOTH (crates/crescent/src/pocket_smooth.rs); on selects $r = 0.75\,\mathrm{mm}$ and $a = 0$. It is off by default, and an enabled setting is part of the certificate identity.

Settings and fitness. We chose the setting on the previous best design, w406t3_r01_p10_c16, exporting it at $r = 0.5$, 0.75, 1.0, 1.5 and $2.0\,\mathrm{mm}$ with $a = 0$, and at $r = 0.75\,\mathrm{mm}$ with $a = 1$. Every export was re-assessed with the complete Stage-2 evaluator, so all twenty objectives, including the STEP solid, are measured on the smoothed geometry; mesh smoothing, by contrast, loses the 5-point CAD/B-rep objective (75.06–75.34 on the ladder). Table IV gives the results together with an unsmoothed re-run of the same evaluator, which reproduces the retained 80.299390 exactly. The current best design was then assessed at the chosen setting only.

TABLE IV

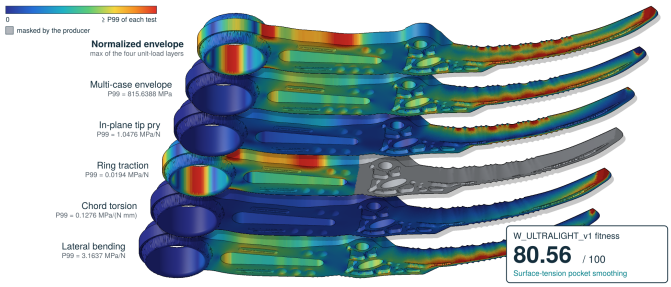
COMPLETE STAGE-2 RE-ASSESSMENT AFTER POCKET SMOOTHING: THE SETTINGS STUDY ON THE PREVIOUS BEST DESIGN, W406T3_R01_P10_C16 (TOP), AND THE CURRENT BEST DESIGN, EC3I042_R01_P10_C56, AT THE CHOSEN SETTING (BOTTOM; ΔF AGAINST ITS OWN AS-EXPORTED 80.623796, BATTERY 13/18). F IS W_ULTRALIGHT_V1 OUT OF 100; THE TIP COLUMNS ARE THE 3D P2 SOLID-TIP SAFETY FACTOR AT $1.82\,\mathrm{N\,m}$ (NODAL MAXIMUM AND AVERAGE); LATERAL IS THE HUNTER CAPACITY AT CERTIFICATION AND STRICT MESHES.

Setting	F	ΔF	Mass (g)	Tip SF	Lateral (N)	Battery
As exported (re-run)	80.299390	–	83.2231	0.6823 / 0.7569	181.5 / 182.2	14/18
$r = 0.5$, $a = 0$	80.164307	−0.135083	83.2178	0.6353 / 0.6645	181.7 / 182.4	15/18
$\boldsymbol{r = 0.75}$, $a = 0$ (chosen)	**80.370405**	**+0.071015**	83.2089	0.7024 / 0.7511	181.8 / 182.5	14/18
$r = 1.0$, $a = 0$	80.308595	+0.009205	83.2070	0.6811 / 0.7237	181.9 / 182.5	14/18
$r = 1.5$, $a = 0$	80.158645	−0.140745	83.1869	0.6275 / 0.6751	182.2 / 182.7	14/18
$r = 2.0$, $a = 0$	78.275029	−2.024361	83.2001	unmeasured	182.3 / 182.8	13/18
$r = 0.75$, $a = 1$	80.161219	−0.138171	83.2629	0.6349 / 0.6752	182.2 / 182.7	14/18
Current best, $\boldsymbol{r = 0.75}$	**80.561227**	**−0.062569**	83.5582	0.6450 / 0.6818	201.1 / 199.9	14/18

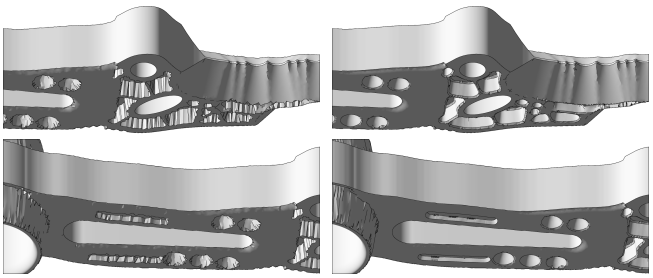
Results. On w406t3_r01_p10_c16, smoothing leaves the engineering responses essentially where they were. The removed volume of every basin is preserved to rounding, and the exported volume changes by $-3.19\,\mathrm{mm}^3$ ($-0.017\,\%$) at $r = 0.75\,\mathrm{mm}$. Mass stays at 83.19–83.26 g against 83.2231 g as exported; the lateral capacity rises by 0.2–0.8 N; the ring, body-twist and fatigue margins move by less than 3 %; the failure-sequence ratio rises from 1.29 to 1.45–1.62; and the CAD/B-rep objective keeps its five points in every case.

What moves F is the 3D solid-tip submodel. Its nodal-maximum safety factor changes non-monotonically with r (Table IV). At $r = 2.0\,\mathrm{mm}$ the tip solve produced no observation, which the evaluator scores as zero. We read these changes as the sensitivity of a nodal maximum near the fuller end to the local triangulation, not as a structural trend (the two solvers agree within $0.08\,\%$). The setting $r = 0.75\,\mathrm{mm}$, chosen visually, is also the best measured on that design.

Larger r merges small pockets into rounded blobs, and the film variant ($a = 1$) spreads depth into the ribs; neither improves F. The chosen export has about 40 % more triangles



(a) Retained plate responses of the smoothed design



(b) Raw export (left) and $r = 0.75\,\mathrm{mm}$ (right): guard, handle

Fig. 4. Surface-tension pocket smoothing of the best design, ec3i042_r01_p10_c56, at $r = 0.75\,\mathrm{mm}$ ($F = 80.561227$). (a) The five retained plate responses of the smoothed design (bottom to top: unit lateral bending, chord torsion, ring traction, in-plane tip pry and the multi-case envelope), each normalized by its own area-weighted 99th percentile, with their normalized envelope on top; colour is a within-test rank, not a safety factor. (b) Guard (isogrid pockets) and handle (flutes, lattice dimples), raw and smoothed, under the ladder's guard camera with $2.5\times$ thickness; the outline, holes and grinds are unchanged, and the pockets shown keep their volume, depth and rib widths.

(STEP 86.7 against $53.4\,\mathrm{MB}$) and a larger tip mesh ($177{,}984$ against $91{,}008$ tetrahedra).

Current best design. At the chosen setting, the design of Figure 1 scores $F = 80.561227$, 0.062569 below as exported: lateral capacity, failure sequence and fatigue gain 0.069 points, and the solid-tip safety factor costs 0.131 (setting not re-tuned). One $0.22\,\mathrm{mm}^3$ sliver of its 15 basins fills in; the rest keep their volume.

Scope. The smoothing changes interior pocket geometry only and adds no qualification or physical observation. The Individual 2000 plate fields of Section VIII are retained evidence of a different design and are not smoothed. Receipts, scorecards and every exploratory setting are in experiments/pocket-surface-tension-smoothing/.

VII. Recorded Search and Inheritance

A. Four Collections with Different Meanings

Births, population memberships, archive representatives, and assessed exports answer different questions. Births record proposed individuals and their observations. Memberships record occupied island slots, including retained individuals. Quality-diversity archives retain representatives under a run's descriptor and score rules. Later assessments concern selectively admitted or separately exported candidates. Evidence coverage and denominators for these collections are reported with the cohort-specific results (Section IX).

For the historical cohort, the recorded membership stage is *after contextual rescoring and before ring migration*. The bound producer emits birth or elite events and summary observations before migration. Migration subsequently copies selected IDs to destination islands; the overwritten destination IDs are not retained by those events. Calling the earlier snapshot a generation-end post-migration population would therefore overstate what can be reconstructed. Elite events retain membership but omit refreshed contextual fitness, so an individual's stored birth score is not substituted for an unrecorded later score.

B. Variation in the Historical Producer

The historical producer uses four island populations, shared novelty memory, a quality-diversity archive, elite retention, tournament selection, and periodic ring migration. Parents can be selected from the archive or current population; recorded parent IDs and operation labels identify the observed birth mechanism. These are source-bound historical semantics, rather than an assertion that every retained cohort executed an identical binary.

Crossover inherits the silhouette's 40 control points as one block and chooses the remaining numeric coordinates and material category independently between parents, then clamps the result. Mutation selects among six modes: sparse coordinate perturbation, a smooth silhouette bump, a thickness-and-section operator, a lattice operator, a fuller/texture/feature operator, and a machining-or-rail operator. Mutation scale depends on island and configuration; the exact parameter branches and the coarse/fine distinction are retained in Supplement I. This is variation within a fixed representation and decoder.

C. Ancestor Reuse in the Retained Ledger

Figure 5 uses the current generated inheritance ledger's three-hop view. Individual 1095 has recorded parents 413 and 526, and 526 itself has parents 413 and 392. Individual 413 is therefore reused along the paths $413 \to 1095$ and $413 \to 526 \to 1095$. The dashed $174 \to 299$ link is a recorded relationship retained as induced closure, distinct from the three-hop edges. The ledger preserves exact coordinate differences and parent ordinals; the main figure emphasizes relationships rather than repeating their raw mixed-unit magnitudes.

This ancestry illustrates a reconstruction capability. Its focal birth's high stored search score and later assessment disagreement are separate observations, developed in Section IX; neither ancestry nor score alone establishes physical performance.

VIII. Score Definitions and Comparability

A search preference, a stored reporting vector, a shortfall score and a mechanical safety factor answer different questions. Their names, directions, observation stages and missing-input rules must accompany their values. Table V keeps the score regimes separate; a shared word such as "fitness" does not create a common performance axis.

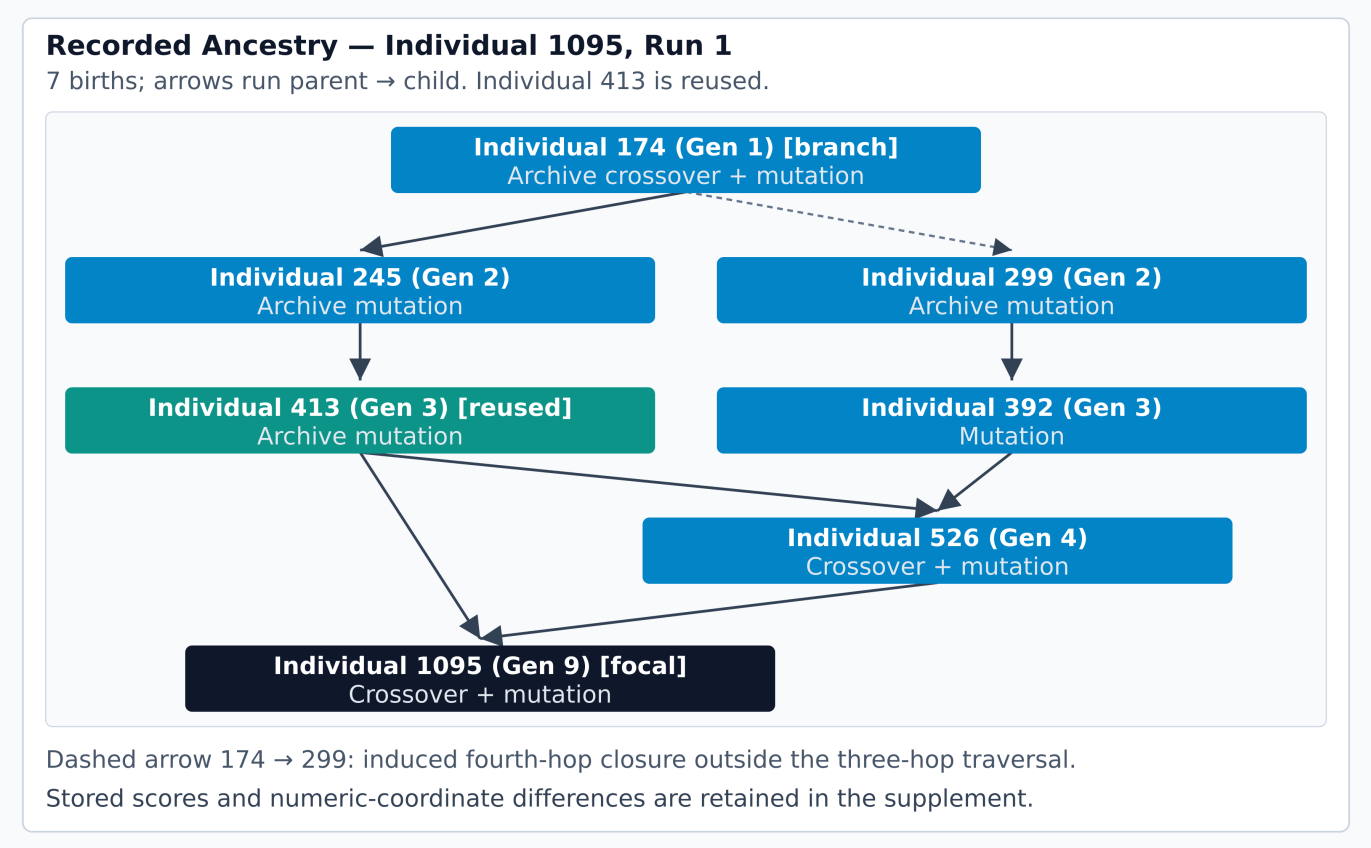


Fig. 5. Shared ancestry in the historical search record around Individual 1095. Solid edges show recorded parent-to-child links within three parent hops; the dashed edge 174 → 299 closes the induced subgraph but lies four hops from the focal individual along that path. Individual 413 contributes through both a direct and an indirect path. The genealogy belongs to this historical cohort.

TABLE V
SCORE DICTIONARY. SCORES ARE DIMENSIONLESS; PHYSICAL UNITS REMAIN ATTACHED TO THEIR INPUTS. THE REPORTING RECONSTRUCTION INVENTS NO MISSING SCORES; HISTORICAL SCORER DEFAULTS ARE DISCLOSED BELOW.

Quantity / stage	Direction / scale	Meaning and normalization	Comparison / missingness
Native f; historical and completed native search	Higher preferred; conditional negative branches, no common $[0, 1]$ scale	Mechanical/mass terms and contextual novelty after legality/feasibility branches	Bound profile, configuration and stage required; novelty changes with context; negative legal scores occur in completed native runs
Screening deficit; intermediate proposals	Lower shortfall preferred; nonnegative stored sum	Stage-1 screening quantity; not the later 20-term reporting vector	Only observed screens have numeric values; 1390 proposals lack screens
Weighted F; retained reporting curves	Higher preferred; $[0, 100]$	Twenty saved normalized terms with weights totaling 100	Compatible versioned recipe recovered; exact executed producer and point-stage comparability unresolved; some inputs are default-compatible
W_{CAD}; 16-gate CAD scorecard	Higher preferred; conditional positive/negative branches	Pass-branch quality versus failed-branch normalized-shortfall penalty	Same assessment definition/inputs required; a favorable subscore is not all-gate success
G_1; assessment composite	Higher preferred; dimensionless	Weighted solid-tip factor, strength/mass, shape, mass penalty and sequence gap	Missing required inputs produce incomplete status; a screen-tip substitute is marked provisional; does not override gates

A. A Retained Weighted Reporting Vector

The principal reporting definition is

$$F = \sum_{i=1}^{20} w_i s_i, \qquad s_i \in [0, 1], \quad \sum_i w_i = 100. \tag{2}$$

Its priorities are mass/compactness (30%), form/comfort proxies (25%), exported geometry (15%) and response/verification (30%). Table VI shows sums of *stored weighted contributions* for Individual 2002. The Supplement retains all twenty normalized values, weights and contributions. These are reported

TABLE VI
INDIVIDUAL 2002: RETAINED REPORTING PRIORITIES AND CONTRIBUTED POINTS. WEIGHTS EXPRESS RESEARCH PREFERENCES; THEY ARE NOT PROBABILITIES OF SAFETY OR QUALIFICATION.

Objective group	Weight (%)	Points
Mass and compactness	30	15.470
Form and comfort proxies	25	23.875
Exported geometry	15	11.841
Response and verification	30	23.022
Total	100	74.207

observations from its summary, not a recalculation under the present scorer and not measurements transferred from Individual 2000.

The saved scalar is $F = 74.207299$, with reported raster mass $101.139\,\mathrm{g}$ and solid-mass scalar $101.1384\,\mathrm{g}$. Its exact genome and certificate link to a retained proposal and screen; a separate leaderboard retains a later-assessment summary with qualified=false. The underlying exported geometry and full scorecard files are not retained at its referenced paths in the checked repository tree. The screen's earlier "pending independent top-k assessment" label is not that later summary's outcome. Individual 2000 supplies the illustrated assessed geometry and response fields.

A versioned twenty-term recipe is recoverable, but its exact historical execution and the curves' point stages remain unresolved. Several saved inputs match that recipe's hard-coded fallbacks for bore radius, edge/point profiles and overload proxies. The four whole terms containing those inputs contribute 15.682353 points; this is not a claim that every point in those terms is an imputed measurement. Some absent-key Boolean branches also receive favorable defaults. The Supplement distinguishes these default-compatible inputs and historical bookkeeping from new reporting-layer imputation. The original vector is preserved, without rescoring.

For the assessed Individual 2000, the actual failed-branch score is $W_{\mathrm{CAD}} = -5.570408$. The same scorecard stores unpenalized quality 1.456983; that is the hypothetical pass-branch value, not an additional achieved score. The Supplement gives the branch equations, input names and shortfall arithmetic. Those scores and G_1 remain separate from F and native f.

B. Context and Display Direction

Native fitness includes novelty computed against contemporaneous population/archive descriptors. A value can therefore change when its context changes, even if the genome does not. Birth scores and later membership observations also have distinct stages. Cross-generation or cross-run maxima describe retained contextual preference; they do not isolate a physical improvement. The historical and newer native implementations/configurations require separate provenance even when both call their scalar f.

A display loss $L = -f$, $100 - F$, or $-W_{\mathrm{CAD}}$ reverses numerical order. Minimizing it preserves the original preference for a larger native score. It does not make those different profiles comparable. A cumulative maximum is monotone by construction, so its corresponding loss is nonincreasing by construction; that fact alone is not evidence of continuing physical improvement.

C. Missing-Aware Response Display and Physical-Unit Views

The display roster accounts for all 44 catalog predicates and 8 retained-channel metadata rows. These 52 metadata rows are not 52 independent tests or performed solves. Four distinct retained unit-response channels of Individual 2000 share the original plate triangles: lateral, in-plane tip pry, ring traction and chord torsion. The scaled envelope and its derived safety-factor map remain companion quantities, without another equal weight. Scalar-only, missing or incompatible-domain observations remain missing spatial contributions; no scalar or Individual 2002 reporting term is broadcast onto this geometry.

Within each eligible channel, the normalization is an ascending area-weighted percentile midrank of valid original response values. All exactly tied values share the midpoint of their pooled area; a constant valid field has rank 0.5. Original source masks apply, while genuine finite zeros remain valid. For cell i, let K_i contain the available normalized channels. The unweighted display index and its support are

$$c_i = \frac{1}{q_i} \sum_{k \in K_i} r_{ki}, \qquad q_i = |K_i| > 0. \tag{3}$$

An all-missing cell has no index and $q_i = 0$, rather than index zero. Contributor identities and coverage $q_i/4$ accompany the index; 44 and 52 are inventory counts, not its support denominator. No candidate-specific minimum/maximum stretch follows the mean. A high index denotes relatively high response within the available channels, not safety, fitness, likelihood or merit.

Physical values, within-channel normalization, the available-channel mean and display contrast are separate quantities. The full-range quantitative companion (Supplement Figure 2) uses a shared flat navy–teal–yellow–orange–red transfer for surface and legend over $[0, 1]$; this palette does not claim monotone luminance. Lighting in the main appearance view changes RGB values without changing the underlying index or retained responses. A percentile-window contrast view is a separate coordinate with declared index bounds, saturation counts and source-area fractions, accompanied by the full-range view. The Supplement specifies the roster, masks, normalization and projection limits.

Figure 7 gives source-coordinate views of the envelope and three physical-unit responses; the fourth eligible response, in-plane tip pry, appears in Figure 8C. Its saliency policy selects a concentrated upper-value region only when whole-field coverage, compactness and scale-gain requirements are met; otherwise it retains full context. B/D display a wider context camera spanning at least 30% of the whole object's horizontal extent, while keeping the selected region's statistics separate. An upper-value cutoff is a fraction of the original eligible value range, not a count or area quantile. Gray denotes producer-masked cells, not measured zero stress. Raw torsion retains its tip-patch peak, distinct from the post-exclusion peak used by selected capacity calculations.

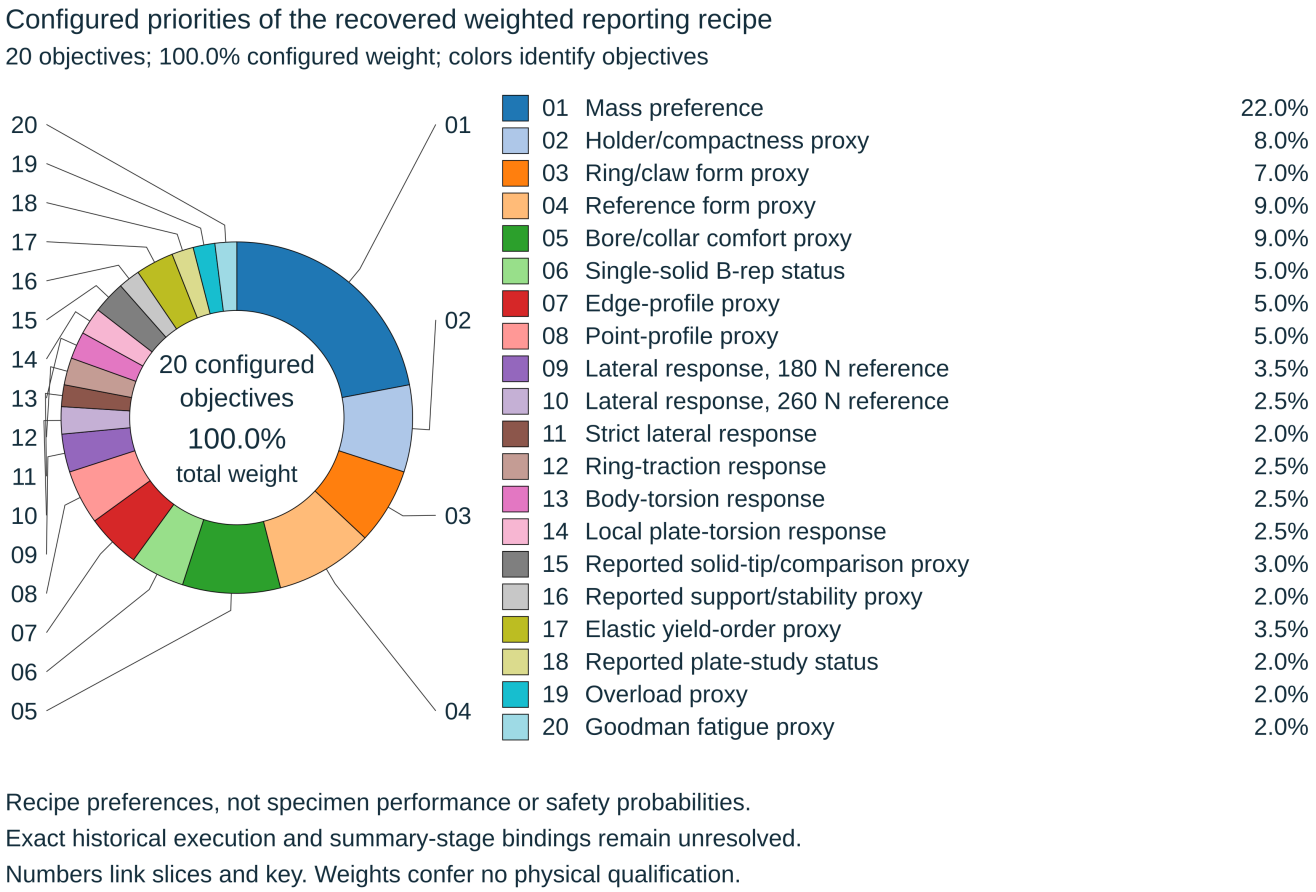


Fig. 6. Configured priorities of the recovered twenty-term reporting recipe: twenty weights total 100%. The chart describes the recipe, not Individual 2000's contributions or Individual 2002's achieved F. It establishes neither gate-free selection nor safety or qualification; historical execution, conditional branches and default-compatible inputs remain separate limitations.

IX. Three Findings from the Retained Records

A. Large Record Coverage Does Not Imply Rich Assessment Coverage

The ten completed native runs (Runs 2–11) retain 1,954,560 evaluated birth records, 195,456 per run across generations 0–800. The admission receipts separately count 32,040 generation/island summary rows. All birth rows preserve the numeric coordinates and material, but no per-birth later CAD/solid assessment is retained for this population. The largest archive therefore supplies search-record coverage rather than a qualification rate.

Figure 9 keeps the native population separate from the historical 1984 births and the intermediate campaign's 2259 numeric screens. The 1390 proposals lacking screens have no invented score. Within the native archive, 754,236 births (38.59%) are search-legal; 610,051 of those have negative contextual fitness. Score sign consequently cannot stand in for legality. Per-run legal counts range from 70,278 to 80,132, with mean 75,424 and sample standard deviation 2937 births. Run-maximum stored fitness ranges from 1.1089 to 1.1440 (mean 1.1250, sample standard deviation 0.01285). These are descriptive summaries of ten selected runs; related births do not provide millions of independent replicates.

Supplement Figure 6 adds directed tail detail: empirical Q90 through the actual retained maximum for native and historical higher-is-better fitness, and the actual minimum through Q10 for lower-is-better screening deficit. Cutoffs use $\lfloor p(n-1) \rfloor$ and include all boundary and favorable-extremum ties. Each panel reports its original-unit bounds and exact below/inside/above counts. The opposite tail remains in the full-range overview; its zero-exclusion claim does not apply to the detail selection. Separate per-run point and retained-best context views for Runs 2–11 reuse the same retained births, without connecting different runs into a shared trajectory. Their axes use each run's declared L and key, rather than the pooled overview scale. These additional views add no observations or evidence of physical convergence.

Figure 10 puts every auto-evolve run and evolution wave since 2026-10-17 on the canonical W_ULTRALIGHT_v1: 351 islands and waves on 4 hosts. Each of the 1761 comparable retained Stage-2 assessments (245 from the 2026-10-17/19 waves, 486 from the 2026-10-18 island campaign, 1030 from the continuous evolution running since 2026-10-21) is re-scored from its own scorecard and RS-v2 receipt; stored leaderboard values are not used. Re-assessing retained genomes with the current Stage-2 evaluator reproduces the 2026-10-17/18

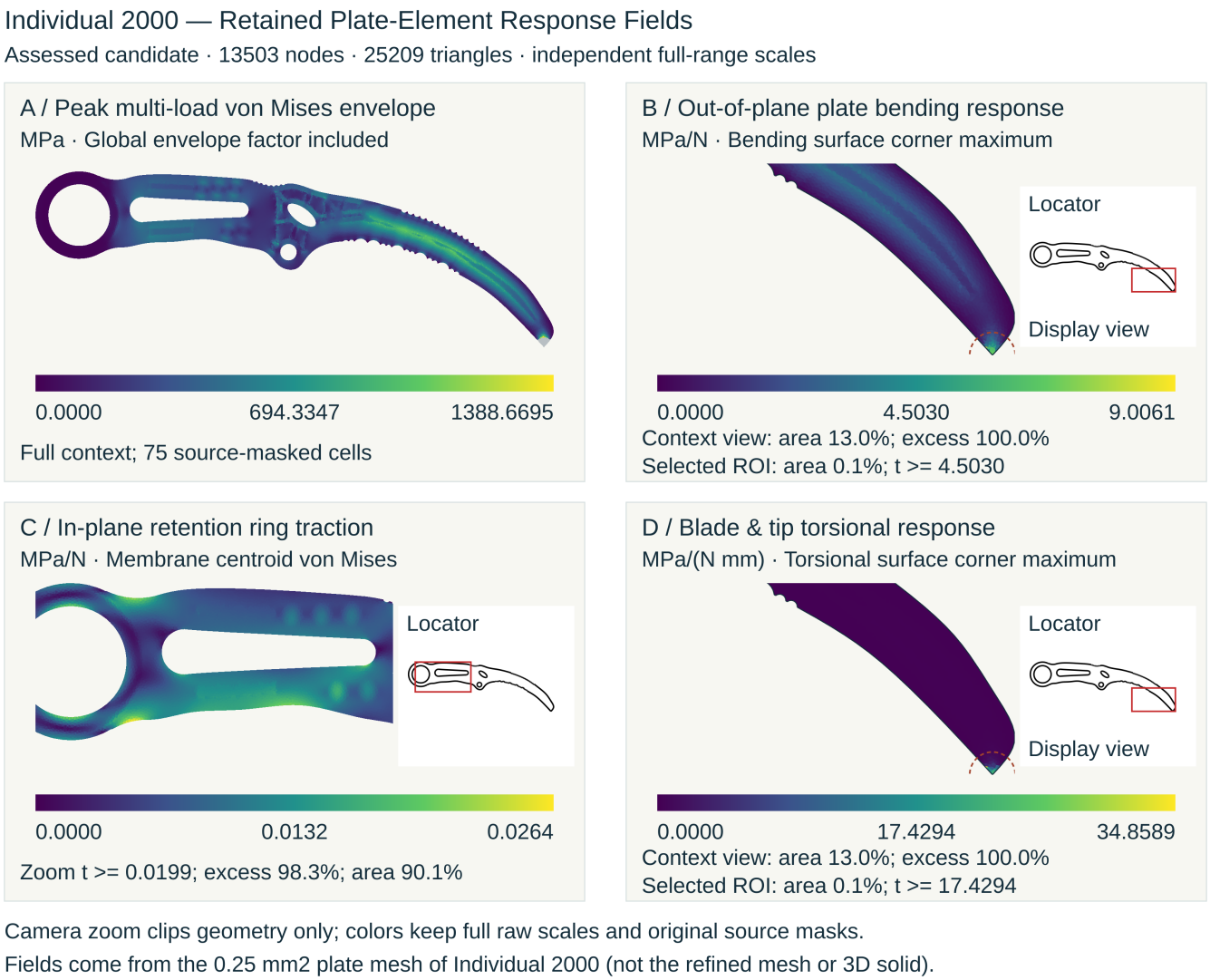


Fig. 7. Source-bound views of Individual 2000's retained plate responses: A, scaled multi-case envelope (MPa); B, unit lateral response (MPa/N); C, unit ring traction (MPa/N); D, raw unit chord torsion (MPa/(N mm)). Each view keeps its full raw color scale, source masks and genuine zeros. B/D widen the camera to show curved blade/tip context; the thin black locator shows the original plate boundary and its red rectangle is the exact displayed viewport. View-area/excess fractions and the smaller selected ROI's area/threshold are reported separately. Coverage uses exact clipped original-triangle areas and each whole eligible field's denominator, not visible ink or local normalization. The source-bound plate-smart-crops.v1.json retains both viewports and statistics. Full context is Supplement Figure 5; in-plane tip pry remains in Figure 8C.

scores exactly (80.299390 and 79.869405), so those waves are comparable, whereas the 2026-10-16 evaluators overstate it (a stored 92.29 re-assesses at 79.166355); their 142 scorecards are counted but not plotted. The 2026-10-18 campaign lost the isogrid pockets, lattice and pierced-fuller windows of the 2026-10-17/18 leader (80.299390); the continuous evolution kept them but, until v2.3.1, forced every child's ring wall to at least 6 mm and peaked at 76.799579. Without that floor it overtook the leader on 2026-10-23 and reached 80.623796 (`ec3i042_r01_p10_c56`, the design on the title page). The right panel summarises 77915 numeric Stage-1 screens as the best screen of each of 766 island rounds; the earlier waves' screens carry no such value. These are search observations under one scorer, not physical tests, and a cumulative best is monotone by construction. The per-birth view of native Run 5 and its definitions moved to Supplement Figure 12.

B. Recorded Ancestry Enables a Later-Assessment Disagreement to Be Inspected

The historical Run 1 retains 1984 births, recorded within-run parent links, 2176 island-slot memberships and 12 migration events. Parent identifiers for external seed ancestry are not recorded. The genealogy exposes reuse of ancestors rather than a simple branching tree. Its saved FEM-status categories distinguish 1455 complete records from 529 not evaluated; the latter are not failed solves.

Individual 1095 has the highest stored historical birth score ($f = 1.15727$, generation 9). Its later strict-assessment summary records cert_pass_all=false and seq_ok=false, although its strict lateral predicate is true. Individual 1543 has the eighth-highest stored birth score ($f = 1.15415$, generation 13), yet its summary records cert_pass_all=true and seq_ok=true. Thus a lower-ranked birth meets the earlier historical certification

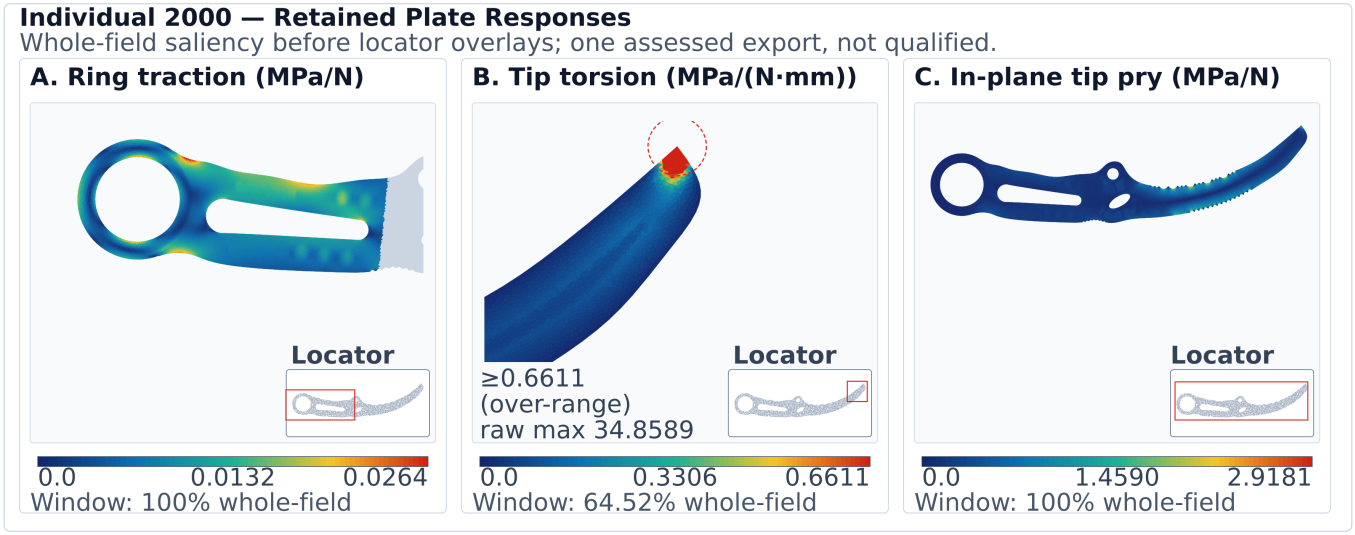


Fig. 8. Individual 2000 retained $0.25\,\mathrm{mm}^2$ plate-field views with whole-object locators. A: ring traction (MPa/N). B: tip chord torsion (MPa/(N mm)). C: in-plane tip-pry T6 centroid response (MPa/N), with stored peak $2.9181\,\mathrm{MPa/N}$. Triangle-intersection integrals of absolute response over the rectangular crop windows retain 100%, 64.52% and 100% of each whole source-active field's integral, respectively. These are physical source-domain fractions before locator overlays, not visible-ink fractions or safety percentages. B marks values $\geq 0.6611\,\mathrm{MPa/(N\,mm)}$ as over-range and discloses the raw peak $34.8589\,\mathrm{MPa/(N\,mm)}$; the capacity exclusion is separate. The Supplement gives the bound domains, masks and denominators.

predicate that the highest-ranked birth fails. This categorical preference–predicate disagreement is not a continuous later-score rank reversal, and that historical predicate is not the later 26-predicate qualification definition.

These are extrema of retained observations under their native preference directions, not newly evaluated designs. The native maximum has seven ties at serialized precision; its earliest original ID and existing public alias are reported without promoting a qualified champion. The source-bound retained-leader receipt preserves every exact tie and each run's own context. Producer/configuration equivalence across the ten native runs is not admitted, including five runs without retained configuration files. Contextual native fitness, screening deficits, weighted reporting summaries and later assessment scores also have different producer and observation-stage bindings. Their values do not establish a single comparable global best. Missing downstream evidence is not a failed assessment, and missing CAD for a score leader does not make the illustrated example a score winner.

The intermediate campaign retains 3649 proposals in 57 birth files across 24 run directories. One-based candidate indexing joins 2259 screens; 1390 screen observations are missing. A separate retained weighted-reporting subset has six curve files and fifteen generation summaries. In Run 31, the archive score changes from 74.1675 (Individual 2009) to 74.2073 (Individual 2002), then stays at that value. An archive maximum is monotone by construction. The curve alone does not show continuing improvement in physical quality. A compatible versioned weighted recipe is recoverable, but the curve does not establish its exact executed producer or uniform observation stage and is not a controlled convergence experiment.

The automatic leaderboard separately retains one assessed export, Individual 2000, with zero qualified and no promoted champion. The historical 23×18 battery includes shared controls, blocked bindings and completeness flags; it is not a set of 414 independent mechanical tests. Individual 2002 also has a retained later-assessment summary: $W_{\mathrm{CAD}} = -7.414831$, $G_1 = 0.259798625$, solid-tip factor 0.3636, 15 battery entries passing and `qualified=false`. Its underlying exported CAD and full scorecard files are not retained at the referenced paths in the checked repository tree. Its linked proposals and screens are retained in another accession. The earlier screen label "pending independent top-k assessment" is not its final retained assessment-summary status. Neither that summary nor its weighted vector supplies the geometry or response fields of Individual 2000.

C. Selected Solver Agreement and Refused Plate Extrapolation Support Different Conclusions

On the same saved solid mesh, the scikit-fem/PyAMG and DOLFINx/PETSc paths agree within 0.05% for the two retained recovered safety-factor observables. Separately, the plate mesh sequence does not admit its selected extrapolated error estimates: one observed order is nonpositive and the other sequence oscillates. Section XI gives the inputs, recovery definitions and refusal reasons. Close shared-input agreement does not repair the extrapolation assumptions or establish physical qualification.

X. What the Numerical Observations Represent

The retained mechanical observations answer different questions. A unit-load plate field, a later assessment scalar, a shared-mesh solver comparison, and a software-control flag do not provide interchangeable evidence. Table VIII groups the eleven observation families by the conclusion each can support. The detailed formulas, source keys, and historical

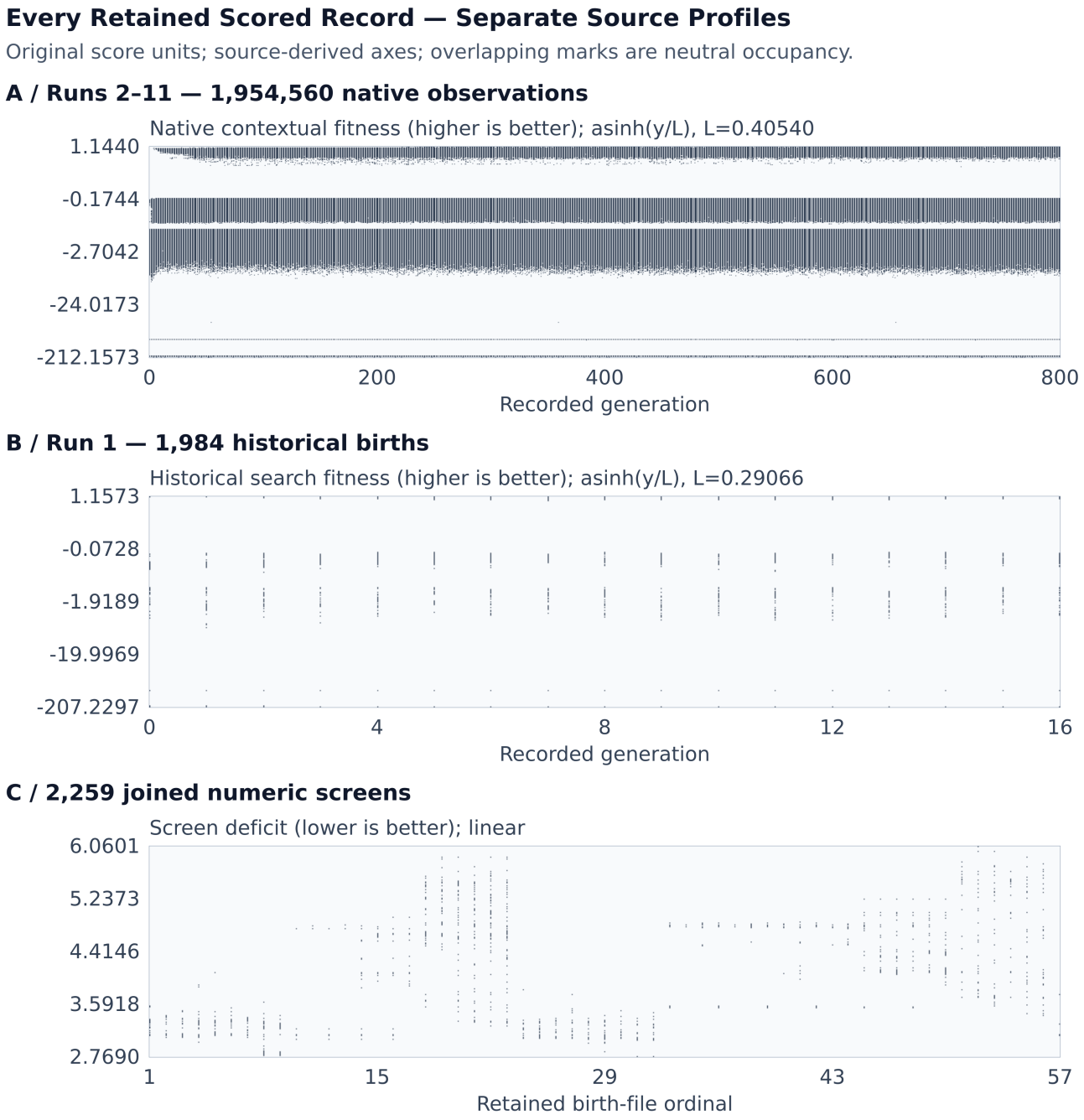


Fig. 9. Full-range overview of separate score profiles: 1,954,560 native births, 1984 historical births and 2259 numeric screens. Every admitted numeric record receives a point operation (1,958,803 total); overlapping marks do not encode density. The pooled native and historical vertical axes use $\operatorname{asinh}(y/L)$ with $L = 0.4054$ and 0.29066, respectively: the empirical 25th percentile of nonzero absolute scores in each profile. Zeros remain plotted. Screening is linear; ticks show original score values and the profiles remain incomparable. The coverage note discloses 1390 unscreened proposals without inventing scores. Zero numeric exclusions applies only to this full-range admitted universe; other campaigns, aborted attempts and unretained records remain outside it.

catalog are retained in the Supplement; the fixed 23-subject historical battery is separate from the single automatically assessed export, Individual 2000.

A. *Plate Fields and Assessment Stages*

For Individual 2000, the retained plate mesh contains 13,503 nodes and 25,209 triangles. Lateral bending and chord torsion use DKT bending; in-plane tip pry and ring traction use quadratic T6 membrane elements. Their unit-response maps

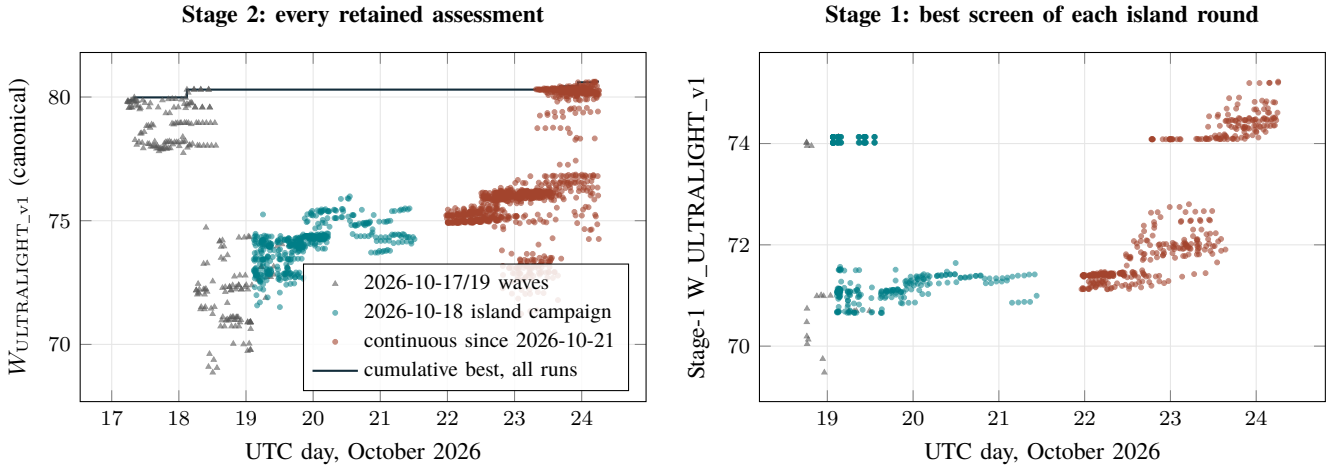


Fig. 10. Every auto-evolve run and evolution wave on every host since 2026-10-17 (351 islands and waves, 4 hosts). Left: all 1761 comparable retained Stage-2 assessments, each re-scored with the canonical W_ULTRALIGHT_v1 from its own scorecard and RS-v2 receipt and placed at the scorecard's file time; the dark step is the best score over all runs so far, ending at 80.623796 (`ec3i042_r01_p10_c56`). 142 scorecards from the 2026-10-16 evaluators, which overstate the score, are not plotted. Right: the best Stage-1 screen score of each of 766 island rounds (77915 numeric screens). Stage-1 and Stage-2 values are different stages (for the same design the Stage-2 value is typically 2–4 points higher) and are not compared. No point is a physical test or a qualification. Data and census: docs/paper/allruns/.

TABLE VII
CANDIDATE ROLES AND SOURCE-BOUND STORED EXTREMA. EACH SCORE RETAINS ITS OWN PROFILE, STAGE AND SOURCE CONTEXT; THE ROWS ARE NOT A COMMON RANKING. A PUBLIC IDENTIFIER OR REGISTRY CATEGORY DOES NOT SUBSTITUTE FOR A RETAINED GEOMETRY OR ASSESSMENT ARTIFACT.

Individual	Role	Retained evidence	Interpretation
1095 / 1543	Run 1 birth ranks 1 / 8: $f = 1.15727/1.15415$	Birth and earlier certification-summary records	Earlier predicate false / true; not later qualification
781909 (Run 5)	Largest numeric stored native $f = 1.1440$; seven ties	Earliest tied original ID 185541, generation 760; retained population row	No bound per-birth later CAD/solid assessment; qualification unknown
2000	Editorially selected retained assessed example; stable alias, not rank	Exact proposal-content link; bound CAD, plate fields and assessment bundle	Parents unrecorded; zero qualified; no champion
2001 / 2007	Separately assessed controlled exports	CAD/scorecard records; $W_{\text{CAD}} = -3.879/-3.965$ and $G_1 = 0.624/0.622$	Grouped strict verification not met
5443	Smallest retained screen deficit 2.769032384488388; one tie	Source-file ordinal 32, candidate 6; proposal and screen	Ordinal is not generation; full CAD/assessment join unestablished
2002 / 2009	Weighted reporting records in Run 31	Stored $F = 74.207299/74.167488$; linked proposals/screens; 2002 has a later unqualified summary	2002's referenced full CAD/scorecard absent; scorer/stage binding unresolved

have distinct units and load definitions (Section XI). The scaled multi-case envelope is derived from those maps; it is not an independent fifth load case.

The assessment comparison in Figure 11C is *certification versus strict*, not a generic coarse/refined pair. Certification takes the minimum capacity over maximum triangle areas 0.25 and $0.125\,\mathrm{mm}^2$, while strict assessment uses $0.0625\,\mathrm{mm}^2$. The saved lateral capacities are 168.90 and $168.18\,\mathrm{N}$; the local tip-torsion factors are 0.524 and 0.529. The displayed fields come from the $0.25\,\mathrm{mm}^2$ level. These stages are distinct from the $4/1/0.25\,\mathrm{mm}^2$ plate mesh study.

The global envelope excludes cells within $4\,\mathrm{mm}$ of the tip. The plate body-torsion observable searches beyond $15\,\mathrm{mm}$ from the tip; the local tip observable searches the 4–$15\,\mathrm{mm}$ annulus. Raw torsion imagery retains the tip-patch response and must distinguish its raw peak from the post-exclusion capacity peak. The ring-case producer masks a half-plane beginning $3\,\mathrm{mm}$ before the seat; it includes the seat and unloaded blade. Those cells are masked rather than interpreted as measured zero stress.

B. Local Solid Response and Recovery

The retained P_2 solid submodel uses $91{,}008$ tetrahedra and $416{,}583$ displacement degrees of freedom. Its cut-arc displacement degrees of freedom are clamped, and vertex forces in a $2\,\mathrm{mm}$ tip patch apply $1.82\,\mathrm{N\,m}$ chord torque. The retained comparison is scikit-fem/PyAMG CG versus DOLFINx/PETSc CG/GAMG on that same saved submodel. Rust orchestration

TABLE VIII
RETAINED OBSERVATION FAMILIES AND THEIR LIMITS. THE ENTRIES DESCRIBE AVAILABLE RECORDS, NOT NEWLY EXECUTED ANALYSES. FIELDS AND ASSESSMENT SCALARS SHOWN IN THE MAIN FIGURES BELONG TO INDIVIDUAL 2000.

Families	Observable	Retained evidence	Supported interpretation
Representation/export; reference shape	B-rep and mesh identity; mass (g); contour distance (mm) and similarity	Bound export records and reference comparison	Inspectable geometry and resemblance to an authored prior; no ergonomic approval
Lateral/pry; ring traction; plate torsion	Unit responses (MPa/N or MPa/(N mm)); capacities (N); regional factors	$0.25\,\mathrm{mm}^2$ cell fields; certification and strict scalars	Elastic responses under named loads and masks; no strict-mesh field is retained
Quadratic solid tip	Recovered vertex stress (MPa) and safety factors	Shared-mesh scalar pairs and distance-bin profiles	Agreement of selected recovered observables; no full solid stress array in the publication inputs
Support/stability; yield-order proxy	Supported regional ratio; unvalidated historical buckling record; elastic capacity ratio	Scalars and a three-level proxy sequence	Dependence on support/region assumptions; no physical fracture sequence
Fatigue	Modified-Goodman factor	Saved scalar $n_f = 0.472$	A model-based cyclic-load margin; no observed fatigue life
Discretization/solver consistency	GCI admission; relative difference of safety factors	Refused plate estimates; solid pairs differing by $< 0.05\%$	Separate evidence about mesh-study admission and shared-input agreement
Software/completeness controls	Contract outcomes, file completeness and score bookkeeping	Historical matrix and control definitions/receipts	Evidence organization and software behavior; no additional mechanical specimens

is not the primary solid assembly implementation. Section XI specifies the common inputs, recovery and selected agreement; the publication inputs contain scalar and binned profiles, rather than a full three-dimensional stress field.

C. Proxies, Support Assumptions and Missing Observations

The regional yield-order ratio compares modeled elastic capacities and has a stored target of 1.2. It does not model plastic redistribution, cracking or a physical failure sequence. Likewise, the modified-Goodman fatigue factor ($n_f = 0.472$) is a scalar model result, not an observed cycle life.

An alternate support study uses bore-bearing springs and a Winkler grip-pad foundation. Its lateral regional ratio (5.203) remains separate from the certification multi-case ratio (1.435); these differing support/load observables are not a mesh-convergence pair. The original buckling value and pass flag are preserved as historical records, but are withdrawn from interpretation as an admitted buckling-performance result. The producer applies a symmetric eigensolver to $\mathbf{K}_s^{-1}\mathbf{G}$ without a retained operator-admissibility check or eigenpair residual. No replacement eigenvalue or rescored fitness is introduced. Figure 11A supplies historical support bookkeeping, not a validated stability margin; no support eigenvector field is retained.

D. Geometry Records and Control Catalogs

Export identities and reference-contour comparisons bind what was generated and measured. They establish neither physical performance nor human comfort. The historical 23×18 battery and 44-entry crosswalk remain available in the Supplement. They mix candidate-specific observations, shared software controls, missing bindings and completeness flags. In particular, its 20 quadratic-solid completeness labels do not denote 20 favorable strength margins. Catalog definitions, executed controls and unavailable records are reported separately; their counts are not a mechanical replication denominator. The stored plate_gci_valid_3level predicate is historical status bookkeeping: its favorable flag does not establish an admitted extrapolated error estimate when the retained GCI procedure refuses the sequences. Original predicate names and outcomes remain unchanged.

XI. NUMERICAL AGREEMENT AND UNRESOLVED ERROR

A. Model, Load and Boundary Identity

The plate and local solid records solve linear-elastic static systems $\mathbf{K}\mathbf{u} = \mathbf{f}$ under their respective geometry, material, support and load assumptions. In the plate path, DKT bending supplies lateral response (MPa/N) and chord-torsion response (MPa/(N mm)); quadratic T6 membrane elements supply in-plane tip-pry and ring-traction response (MPa/N). Lateral and torsion values are per-triangle maxima over three corner-extrapolated bending stresses; pry and ring values are T6 centroid von Mises stresses. The envelope takes the maximum of lateral and pry responses scaled by $260\,\mathrm{N}$, ring traction scaled by $1500\,\mathrm{N}$, and torsion scaled by $1820\,\mathrm{N\,mm}$ with the producer's global torsion correction. That factor is selected at a peak section and applied globally; it is an approximation rather than a locally recovered three-dimensional torsion field.

The local solid model uses quadratic (P_2) displacement fields on linear tetrahedral geometry. It clamps cut-arc vertices and edge-midpoint degrees of freedom to zero. Minimum-norm out-of-plane vertex forces on the $2\,\mathrm{mm}$ tip patch supply $1820\,\mathrm{N\,mm}$ chord torque, with zero resultant force and zero transverse moment. The retained producer does not transfer displacements from a larger model. This cut-arc boundary idealization is shared by both solid implementations.



Fig. 11. Individual 2000: retained scalar assessment observations and conditional score bookkeeping. A: support assumptions and historical scalars; the buckling record is not admitted as a stability-performance result. B: modified-Goodman factor for the modeled 0–180 N lateral cycle and overload capacity/reference-load factors: 16.2 N m torsion, 1000 N in-plane tip (record only), 3427 N spine and 3000 N ring-back. Larger factors mean greater modeled margin; unity is the named-load reference, not overall qualification. C: certification minimum over $0.25/0.125\,\mathrm{mm}^2$ versus strict $0.0625\,\mathrm{mm}^2$ observations: lateral capacity 168.90 versus 168.18 N and local tip-torsion factor 0.524 versus 0.529 relative to 1.82 N m. D: continuous CAD subscores and shortfalls for this assessed candidate, separate from the retained weighted reporting vector of Individual 2002. This is one assessed automatic export, with no qualified or promoted champion in its retained leaderboard.

B. Field Support and Composite Interpretation

For the 25,209 retained plate triangles, the envelope has 25,134 active cells and 75 producer-excluded cells within 4 mm of the boundary tip node. Its saved active peak is 1388.67 MPa. Ring traction has 10,334 active cells and 14,875 producer-zeroed cells in its source-masked half-plane, which extends beyond the seat. The latter are masked; their zeros are not stress measurements. Lateral, tip-pry and raw torsion maps retain all 25,209 cells, including the tip region that is excluded from selected capacity calculations. Source-coordinate rounding makes one ring-boundary cell ambiguous geometrically; its positive stored ring response establishes that the producer retained it.

The response display averages pooled area-weighted per-

centile ranks over four available channels, with original source masks and genuine finite zeros kept distinct. The derived envelope receives no additional equal weight, and the available-channel mean has no candidate-specific minimum/maximum stretch. Per-cell contributor records and source-area coverage use four eligible channels as the support denominator; the catalog and metadata inventories are not that denominator. Supplement Figure 3 shows that support. These dimensionless values locate relatively high responses within the retained fields, rather than safety, fitness or quality. Section VIII and the Supplement distinguish the current display definition from its superseded five-map recipe.

C. Plate Mesh Study: Refused Error Estimates

The retained three-level study uses maximum triangle areas 4, 1 and $0.25\,\mathrm{mm}^2$, corresponding to the nominal mesh-size convention $h = \sqrt{A_{\max}} = 2$, 1 and $0.5\,\mathrm{mm}$ and 3166, 6803 and 25,209 elements. This h is defined from the requested maximum area, rather than a measured element-size distribution. Its stored mesh/observable metadata are valid, but the selected relative-GCI procedure [12] admits neither error estimate:

- The yield-order proxy is monotone across the stored sequence but has observed order $p = -2.9003$; the procedure refuses a nonpositive order.
- The lateral safety-factor sequence has oscillatory differences and is refused on that basis.

The stored extrapolated values and fine-grid GCI estimates remain null. This is a failure to establish the assumptions of this retained procedure for these sequences, not a claim that every numerical-error method is impossible. The certification minimum over $0.25/0.125\,\mathrm{mm}^2$ and strict $0.0625\,\mathrm{mm}^2$ scalars are separate assessment stages; they do not supply this three-level study with an additional observation.

D. Shared-Mesh Solid Agreement

The primary solid path assembles scikit-fem vector P_2 tetrahedra [8] and uses PyAMG smoothed aggregation with CG. The comparison path assembles a DOLFINx 0.9.0 vector Lagrange P_2 space [9] and uses PETSc 3.22.0 CG/GAMG. Both consume the same saved mesh, elastic constants, cut clamp and patch forces. A retained runtime capture lists scikit-fem 12.0.2 and PyAMG 5.3.0; the DOLFINx sidecar records its environment. The capture is written before the orchestration may reuse valid cached solid outputs, and the original solid/verification records do not bind its digest. It therefore documents a captured environment, without proving the exact environment of a cached historical solve. The Supplement gives the full captured versions and image identity separately from the assembly and solver records.

Both recover von Mises stress at element vertices and form the incident-cell maximum or unweighted mean. The comparison region is $4 \le d_{\mathrm{tip}} < 15\,\mathrm{mm}$ with distance from the saved cut vertices $> 3\,\mathrm{mm}$. The saved primary safety factors have three decimal places (0.669, 0.726), whereas the comparison has four (0.6693, 0.7263), under maximum and mean recovery respectively. With $\delta = |\mathrm{SF}_{\mathrm{DOLFINx}}/\mathrm{SF}_{\text{scikit-fem}} - 1|$, those stored rounded scalars give differences of 0.0448% and 0.0413%. These are comparisons of the reported safety-factor tokens, not bounds on unrounded agreement or physical accuracy. The primary record reports 152 iterations and relative residual 8.80×10^{-11}; the comparison reports 158 iterations.

This comparison cross-checks two assembly/solver implementations through selected recovered scalar observables under shared inputs. Common mesh preparation, material, boundary and load assumptions can carry common-mode errors. The comparison does not verify every algebraic operation, bound discretization or modeling error, establish mesh independence, or validate the clamped submodel against physical measurements. The retained profile bins use different plate-cell and recovered-solid-vertex observables and therefore require their own interpretation.

E. Open Eigenproblem and Assurance Definitions

The support study's eigenvalue is not admitted here as a current engineering buckling result. Its producer invokes a symmetric eigensolver on $\mathbf{K}_s^{-1}\mathbf{G}$. Symmetric component matrices alone do not establish Euclidean self-adjointness of that product, and no confirming operator-symmetry check or eigenpair residual is retained. The eigensolver's operator contract [11] identifies the unresolved question; it does not supply a historical validation result. The Supplement preserves the original scalar and status as historical provenance. Any qualification inference that depends on this eigenvalue remains unestablished; this reporting revision neither replaces the value nor rescores the historical observations.

The frozen assurance catalog defines 178 challenges, or 356 paired positive/negative control slots across 20 families. Those are definitions, not proof that all slots executed. Individual execution receipts and outcomes must establish which controls ran; unavailable receipts remain unavailable. Neither catalog size nor repeated software-control labels supplies additional candidate-specific mechanical evidence.

XII. Reconstruction and Artifact Access

A. Four Reproduction Levels

Table IX defines these four levels and their evidence boundaries.

B. Recovery Status

Fresh-process resumption of the native search is unproven in this accession. Serializing a checkpoint or continuing in its original process does not establish disk-only reconstruction of the complete running state in a new process. Separately scoped neutral integer-stream recovery work concerns only its own fixture state, accepted journal prefix and storage protocol; it does not establish resumption of a production campaign. Any recovery claim requires an actual fresh-process receipt binding the fixture source/binary, full restored state, journal sequence and uninterrupted comparison. Such a receipt is distinct from report reconstruction and from a new scientific evaluation.

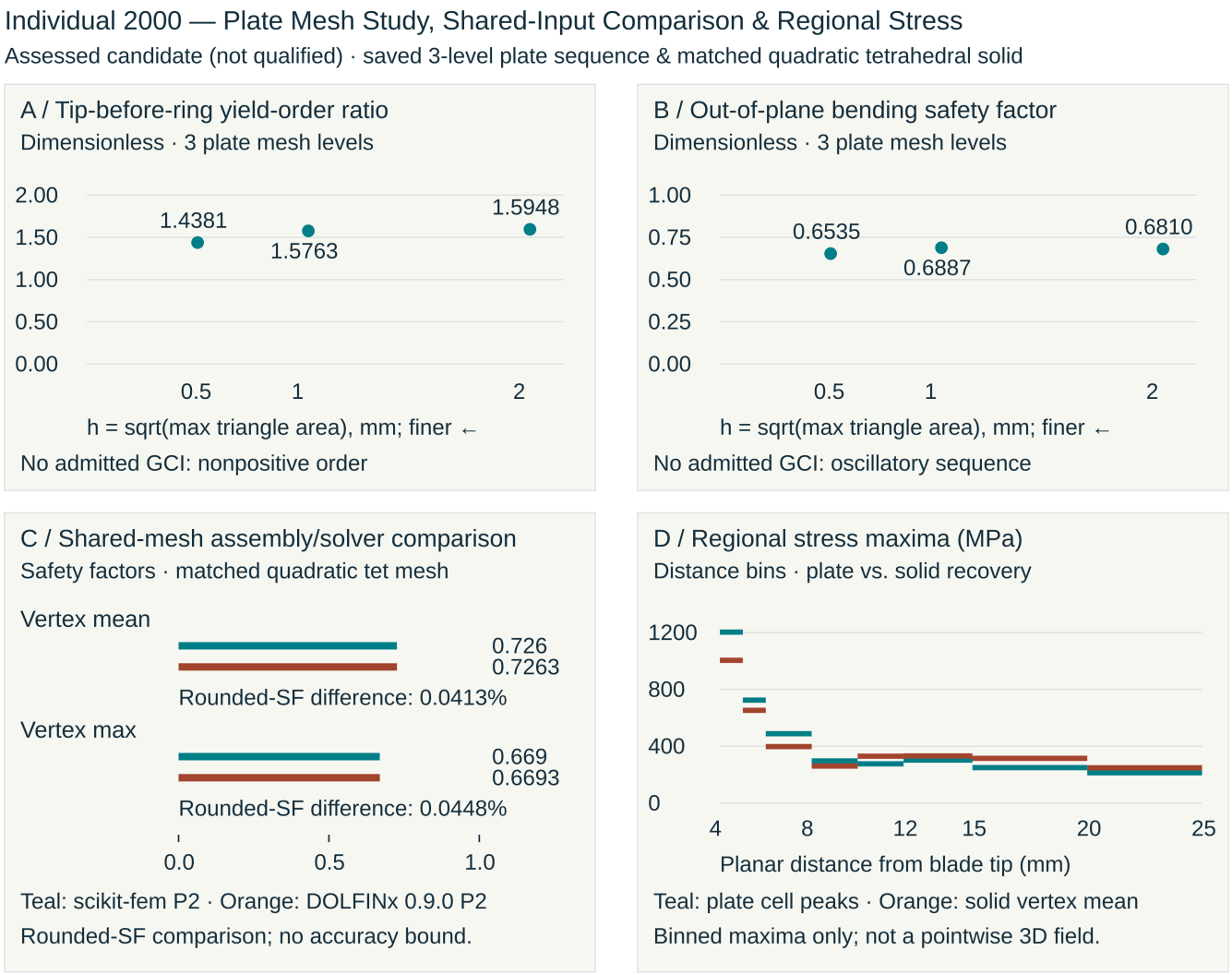


Fig. 12. Individual 2000 retained numerical observations. A–B: plate sequences with maximum triangle areas $4/1/0.25\,\mathrm{mm}^2$; the stored relative-GCI estimates are refused for nonpositive order and oscillatory differences. C: scikit-fem/PyAMG and DOLFINx/PETSc safety factors on the same 91,008-tetrahedron, 416,583-degree-of-freedom solid submodel, using incident-cell vertex maximum and unweighted mean recovery; primary safety factors are stored to three decimal places and comparison values to four. D: distance-bin stress maxima for raw plate cells and recovered solid vertex means; the differently recovered profiles are not a pointwise field-error estimate.

C. Immutable Evidence Entry Points

The retained technical crosswalk provides the public-ID mappings and access to the historical matrix, export identities and original field inventory. The completed-population archive index identifies ten run manifests and 200 compressed segments. These links identify the prior retained evidence accession; they are not a claim that the present revised manuscript was authored at that revision. The final publication receipt supplies the matching manuscript/build accession and standalone supplement access. A TeX include fragment is source, not a compiled supplement download.

Candidate, producer, reporting-source and publication identities remain distinct. For example, a certificate hash binds content identity under its scheme; it does not prove a CAD artifact exists. The archived producer revision is preserved with its original inputs rather than replaced by the present manuscript revision. The matching publication receipt must bind the actual manuscript head/tree, build environment, main and supplement PDF hashes, generated outputs and receiving-main/release accession. Receiving-tree equality establishes the delivered source bytes; it does not reconstruct an unrecorded historical producer execution. Hash equality establishes byte identity, not correctness of the original computation.

D. Reporting Commands and Writes

The reporting environment requires Rust/Cargo with the locked dependencies and complete admitted archives; document production additionally requires LuaLaTeX/IEEEtran,

TABLE IX
DISTINCT REPRODUCTION CLAIMS AND THEIR SCOPE IN THIS STUDY. THE PUBLICATION/RELEASE RECEIPT IDENTIFIES ACTUAL COMMAND OUTCOMES FOR ITS EXACT SOURCE AND OUTPUTS.

Level	Operation	Evidence boundary
Report reconstruction	Read retained contracts/archives; check identities; regenerate reporting views and documents	The publication operation; preserves evidence inputs but writes derived/build outputs
Solver replay	Rerun a bound CAD/FEM workflow under a declared environment	Not performed by the reporting rebuild
Software controls	Execute named positive/negative fixtures and admission checks	Actual receipts establish executed cases; catalog definitions alone do not
Physical replication	Fabricate specimens and obtain mechanical observations	Physical observations are not reported in this study

the declared fonts, librsvg and PDF inspection tools. From the repository root, the following commands request a full compressed-population audit and retain the existing document-build/check entry points. Commands are shown at full width so flags remain copyable; actual exits/toolchain/output hashes belong to the final validation receipt.

The artifact audit decodes admitted historical frames, checks proposal/screen joins, verifies compressed-segment and reporting bindings, and checks generated-output consistency. Its flag-controlled compressed-segment decoder defaults to one segment per run; `--full-zstd` removes that limit. For the admitted completed-population accession, full scope is 200 segments containing $1{,}954{,}560$ rows. The final exact-head receipt must establish the requested mode, actual segment/row counts and successful exit; documenting the command alone does not establish its execution. The document check renders figures and compiles the paper under `target/ieee/paper/`, and the existing build also updates `docs/paper/main.pdf`. Evidence inputs remain read only; the entire operation is not filesystem read only. The supplement target, complete input manifest, all-page visual inspection and clickable-link checks are part of the matching publication package rather than implied by the existence of source files.

XIII. PROSPECTIVE REPRESENTATION RESEARCH

FunSearch provides context for separating language-model program proposals from execution-based evaluation in mathematical and algorithmic settings [6]. It does not establish an agent result in Jarambit. Searching values in the present schema and proposing a different schema or decoder are distinct activities; the retained runs demonstrate the former.

A future controlled study on benign reference CAD tasks could compare human-authored and agent-assisted representation proposals under fixed evaluator versions and review criteria. A proposal would identify types, units, dependencies, serialization and migration rules, and expected tests; retained rejected proposals would permit measurement of invalidity, reconstruction success, and reviewer effort. Those outcomes remain unmeasured here. Human responsibility for the problem definition and interpretation remains essential, while independently controlled evaluators constrain which proposed effects are actually observed.

XIV. DISCUSSION

A. What the Connections Establish

The case study makes three relationships inspectable. Ancestry records connect a birth to recorded sources of variation; candidate and stage identities connect a search preference to a particular assessment; source artifacts connect displayed quantities to the models and transformations that produced them. The usefulness lies in being able to follow these relationships for a concrete result, rather than inferring them from a final geometry or score.

The historical comparison illustrates why the assessment predicate must travel with an outcome. A lower-ranked birth can meet the run's recorded predicate when the highest-ranked birth does not; that observation is meaningful within its stated selection and assessment context. The numerical comparison has an analogous structure: agreement on a shared solid discretization supports consistency of selected recovered observables, while the plate refinement study addresses admission of a different error estimate. Preserving both observations makes their respective implications visible.

B. Scope and Limits

This is a retrospective study of one repository and selected retained workflows. Prior selection and retention determine which observations are available, and missingness is not assumed random. Births share ancestry and search context; numerical observations can share geometry, preprocessing and model assumptions. Counts therefore describe record coverage, rather than independent experimental replication. Contextual fitness and cumulative best-so-far curves describe the recorded preference; neither those curves nor selected assessment comparisons estimate a population-wide qualification rate or establish comparative search advantage.

The mechanical observations are computational model outputs. Shared-input solver agreement leaves discretization, loading and boundary idealizations to their own checks; the support-study buckling estimate is not admitted because its operator/eigenpair assumptions remain unverified. Physical observations are not reported. Material labels and CAD dimensions also leave manufacturing state and process capability to measurement, as illustrated by NIST's test-artifact work [10]. A visible motif and an authored feature family establish neither its autonomous invention nor an isolated engineering benefit. Those questions require additional evidence.

```
cargo run --release --locked \
  --manifest-path crates/paper-artifacts/Cargo.toml \
  --example publication_v2_audit -- --full-zstd
sh docs/paper/reproduce.sh build
sh docs/paper/reproduce.sh check
```

Fig. 13. Reporting entry points, without invoking a new search, CAD export or mechanical solve. The explicit `--full-zstd` flag requests decoding of every compressed segment. The artifact audit reads retained inputs and checks reporting consistency. The Makefile compiles figures/PDFs and writes build products.

C. A Testable Research Direction

The next question is whether explicit links improve reconstruction and interpretation on benign scientific workflows. Controlled provenance defects could test detection and explanation, and independent readers could measure reconstruction accuracy, time and uncertainty. Such comparisons would evaluate the usefulness of the reporting approach itself. The present case supplies a concrete artifact and specific questions for that evaluation.

XV. Conclusion

Jarambit provides an inspectable account of representation, recorded search and selected numerical assessment. Its cohorts expose uneven evidence coverage, a historical contrast between search rank and a recorded assessment predicate, and different conclusions from shared-mesh solid agreement and refused plate extrapolation. These links make the reported associations between candidates and evidence inspectable.

As design searches generate more records, a useful scientific deliverable includes the context needed to interpret them: which representation, which ranking rule, which assessment stage and which original observation. Jarambit demonstrates that account for one retained computational study. Engineered Intelligence joins scalable proposal with engineering evidence whose scope and source another researcher can inspect.

Acknowledgment and Assistance Disclosure

This draft was prepared with AI assistance for source analysis, manuscript organization and prose, documentation tooling, and author adversarial review. Human author approval, contribution verification, independent scientific review, and venue-specific disclosure remain required before submission. No institutional affiliation or funding is asserted.

References

[1] Jarambit project, "Publication source and retained-evidence snapshot," revision 102cdbef52b3710ce360ac34475c96ede0e33970, 2026. [Online]. Available: https://github.com/neverhuman/jarambit/tree/102cdbef52b3710ce360ac34475c96ede0e33970.

[2] R. Wu, C. Xiao, and C. Zheng, "DeepCAD: A deep generative network for computer-aided design models," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 6772–6782. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2021/html/Wu_DeepCAD_A_Deep_Generative_Network_for_Computer-Aided_Design_Models_ICCV_2021_paper.html.

[3] M. S. Khan, S. Sinha, T. U. Sheikh, D. Stricker, S. A. Ali, and M. Z. Afzal, "Text2CAD: Generating sequential CAD designs from beginner-to-expert level text prompts," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 7552–7579, doi: 10.52202/079017-0242. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/hash/0e5b96f97c1813bb75f6c28532c2ecc7-Abstract-Conference.html.

[4] J. D. Lohn, D. S. Linden, G. S. Hornby, W. F. Kraus, A. Rodríguez-Arroyo, and S. E. Seufert, "Evolutionary design of an X-band antenna for NASA's Space Technology 5 mission," NASA Technical Reports Server, document 20030067398, 2003, conference preprint. [Online]. Available: https://ntrs.nasa.gov/citations/20030067398.

[5] J.-B. Mouret and J. Clune, "Illuminating search spaces by mapping elites," arXiv:1504.04909, 2015. [Online]. Available: https://arxiv.org/abs/1504.04909.

[6] B. Romera-Paredes *et al.*, "Mathematical discoveries from program search with large language models," *Nature*, vol. 625, pp. 468–475, 2024, doi: 10.1038/s41586-023-06924-6. [Online]. Available: https://doi.org/10.1038/s41586-023-06924-6.

[7] NASA, *Standard for Models and Simulations*, NASA-STD-7009B, Mar. 5, 2024. [Online]. Available: https://standards.nasa.gov/standard/NASA/NASA-STD-7009.

[8] T. Gustafsson and G. D. McBain, "scikit-fem: A Python package for finite element assembly," *J. Open Source Softw.*, vol. 5, no. 52, p. 2369, 2020, doi: 10.21105/joss.02369. [Online]. Available: https://joss.theoj.org/papers/10.21105/joss.02369.

[9] I. A. Baratta *et al.*, "DOLFINx: The next generation FEniCS problem solving environment," preprint, version 1, 2023, doi: 10.5281/zenodo.10447666. [Online]. Available: https://zenodo.org/records/10447666.

[10] S. Moylan, J. Slotwinski, A. Cooke, K. Jurrens, and M. A. Donmez, "An additive manufacturing test artifact," *J. Res. Natl. Inst. Stand. Technol.*, vol. 119, pp. 429–459, 2014, doi: 10.6028/jres.119.017. [Online]. Available: https://www.nist.gov/publications/additive-manufacturing-test-artifact.

[11] SciPy developers, "scipy.sparse.linalg.eigsh," API reference, accessed Oct. 20, 2026. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.sparse.linalg.eigsh.html.

[12] P. J. Roache, "Perspective: A method for uniform reporting of grid refinement studies," *J. Fluids Eng.*, vol. 116, no. 3, pp. 405–413, 1994, doi: 10.1115/1.2910291. [Online]. Available: https://doi.org/10.1115/1.2910291.

[13] O. Sigmund and K. Maute, "Topology optimization approaches: A comparative review," *Struct. Multidiscipl. Optim.*, vol. 48, no. 6, pp. 1031–1055, 2013, doi: 10.1007/s00158-013-0978-6.

[14] L. Moreau and P. Missier, Eds., *PROV-DM: The PROV Data Model*, W3C Recommendation, Apr. 30, 2013. [Online]. Available: https://www.w3.org/TR/2013/REC-prov-dm-20130430/.

[15] M. D. Wilkinson *et al.*, "The FAIR Guiding Principles for scientific data management and stewardship," *Sci. Data*, vol. 3, Art. no. 160018, 2016, doi: 10.1038/sdata.2016.18.

[16] A. M. Smith, D. S. Katz, K. E. Niemeyer, and FORCE11 Software Citation Working Group, "Software citation principles," *PeerJ Comput. Sci.*, vol. 2, e86, 2016, doi: 10.7717/peerj-cs.86.

[17] F. Crameri, G. E. Shephard, and P. J. Heron, "The misuse of colour in science communication," *Nat. Commun.*, vol. 11, Art. no. 5444, 2020, doi: 10.1038/s41467-020-19160-7.

[18] S. P. Huber *et al.*, AiiDA 1.0, a scalable computational infrastructure for automated reproducible workflows and data provenance, *Sci. Data*, vol. 7, Art. no. 300, 2020, doi: 10.1038/s41597-020-00638-4. [Online]. Available: https://doi.org/10.1038/s41597-020-00638-4.

[19] S. Leo *et al.*, Recording provenance of workflow runs with RO-Crate, *PLoS ONE*, vol. 19, no. 9, e0309210, 2024, doi: 10.1371/journal.pone.0309210. [Online]. Available: https://doi.org/10.1371/journal.pone.0309210.